



Andrew Ibsen

TOKENIZING AIRCRAFT ASSETS ON A DISTRIBUTED LEDGER

An inventory management solution built for British Airways Engineering.

Submitted as part of the requirement for the award of
MSc in Aircraft Maintenance Management
At City, University of London

I certify that this project is wholly my own work and in accordance with the project regulations.

All material that has been extracted from others has been clearly referenced.

This Project complies with the Project Regulation and Guidelines.

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Supervisor: Dr. Paulo Nunes

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Executive Summary

Blockchain technology has the potential to revolutionize inventory management by providing a secure and transparent way to track and manage inventory (Henry and Kathawate, 2023). By tokenizing inventory items on a distributed ledger, companies can increase accuracy and security of inventory management operations (Bhushan et al., 2023). Thus, it can be hypothesized that implementing a blockchain framework for inventory management in an airline's engineering operation will increase accuracy of part's locations and facilitate maintenance tracking, promoting cost-control. This hypothesis was tested within British Airways Engineering operation at John F. Kennedy Airport. The prototyped solution was developed after analysis of the current system and ongoing consultation of potential users. Initial surveys highlighted moderate levels of satisfaction with the existing system with pain points around item retrieval effectiveness. To leverage the airline's existing secure online environment, a Microsoft Power Apps application was built as a front-end to transact inventory items as it is receipted in and out of storage. The application is a gateway to assets listed with unique identifiers on Microsoft SharePoint. Receipting in the mobile application is replicated in a blockchain (Sepolia Testnet) component. The prototype was deployed to 25 engineers and its use tracked through a logging system. Results reveal moderate user acceptance with moderate-to-good usability and reports of improvement on efficacy of item retrieval. Significant improvements in average accuracy ($p < 0.05$) and time spent transacting ($p < 0.05$) were obtained when systematically testing the new solution against the legacy system. Spot checks found a significant decrease in misreported parts locations, saving an estimated \$150'000 to the operation after the first 3 weeks of deployment only. Thus, blockchain has improved supply chain transparency and traceability, potentially prevented loss of parts and increased efficacy of item retrieval. This, in turn, is expected to reduce operational costs for the line maintenance station and increase aircraft turnover rates. Furthermore, the error-proof data collected could be monitored periodically to optimize stockholdings and further benefit the operation. Because the solution is low-cost and functions within the airline's secure environments, it can easily be scaled to larger BA operations.

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1 Introduction

1.1 Context

In 2024, an Alaskan Airline flight lost, in mid-air, a door plug. In the aftermath, the Federal Aviation Authority (FAA) grounded all Boeing 737 Max-9 aircraft in America. Multiple flight cancellations ensued. Alaskan Airlines reported a related cost of 150 million USD. United Airlines, having 79 Max-9 in its fleet, also attributed to this event falling short on their projected quarterly earnings. The ongoing FAA investigation centers on missing bolts meant to secure the door plug in place. Likewise, in 2019, 300 Boeing's 737 jets were fitted with faulty wing components, engendering consequential losses for all owner airlines. At the time, the FAA proposed a \$3.9 million civil penalty against Boeing, highlighting the potential vulnerabilities from third-tier suppliers to final assembly as well as certification, and underscoring the critical need for enhanced traceability for quality control in the aviation supply chain (FAA, 2019). Aircraft are ever-more complex machines that, due to their intense commercial use, incur heavy wear and tear. To maintain high safety standards, each component, from bolts to auto-flight computers, must be thoroughly inspected and properly fitted. Thus, manufacturers and commercial aviation operations must continuously demonstrate high technical knowledge and efficient aircraft maintenance procedures.

In commercial aviation, engineering also has the mandate of ensuring standards within the cabin to paying passengers. The most common demands stem from faults in seats or in-flight entertainment systems, which are central to an airline's hospitality and profitability. In fact, seats are a perishable good, in the words of Abriani, 2019, "you have only one chance to sell a seat prior to departure". Failure in timely maintenance causes seats to be locked out, especially in premium segments such as business and first-class (Nast, 2023) even though those segments, accommodating only 12% of travelers generate over 25% of revenue (Brock, 2022). Not being able to quickly identify if items are available in stores and install them during a transit is thus an important factor in lost revenue due to unsellable seats.

Furthermore, maintenance delays often conflict with crew duty time limits. Regulatory agencies determine maximum amount of time spent in air v. ground time for crews. Once crews are onboard, even if the aircraft has not departed, time starts to be counted against maximum time spent in air. Thus, flights are regularly canceled because of prolonged emergency maintenance conducted while the crew is onboard. As such, difficulty retrieving necessary parts from storage is a contentious point. For instance, recently at John F. Kennedy (JFK), Airport an Iberia aircraft had a malfunctioning CF6 Engine Starter at departure. The engineers on the ground soon realized that the Iberia stock at JFK did not hold a spare starter. This was communicated to Iberia, which sourced a starter from partnering operators at JFK Airport in an hours' time. Unfortunately, when adequate maintenance was eventually completed, the pilots had run out of duty time. Thus, despite successful engineering processes and aircraft readiness, the flight was cancelled. A posteriori, a starter was discovered in stock at British Airway's store's facility. The hour spent sourcing a new starter could have been saved and a \$1 million (iagcargo, 2018) loss due to an Aircraft On Ground

(AOG) situation, avoided. The anecdote highlights how inventory lists must be up to date, centralized, and readily available, to fit the time constraints of commercial aviation.

Beyond the importance of time efficacy for part retrieval, misplaced parts are another pitfall. Lost parts are a regular but rarely divulged occurrence in airline operations, with recurrent monetary impact. For the most part, airlines do not have systematic report systems in place to monitor such events. In spring 2023, BA Engineering, serving its mandate as Iberia's maintenance provider at JFK Airport, removed a faulty Integrated Drive Generator (IDG) from an Iberia aircraft and installed a new part. In summer 2023, Iberia contacted British Airways enquiring about the IDG that had been removed and was believed to be in the Iberia's stockholding maintained by BA at JFK. BA Engineering, upon consulting maintenance records, reported that the IDG had been sent back to Iberia in Madrid. Contradictory paperwork had been filed by both airlines regarding the location of this IDG. The conflict was only resolved when, months later, an unaccounted for and unmarked box was uncovered by chance at a shipping warehouse facility in JFK Airport. The part in question is worth, new, \$ 250'000, and circa \$175'000 used (Aerobay 2023).

Perhaps more frequent than all together lost parts, are arrival of parts with incorrect paperwork. The lack of documentation causes the parts to be returned to the sender, causing stockholdings to be ill-equipped for rapid aircraft turnover. Lack of paperwork, beyond delays, raises a grave safety question. In fact, counterfeit parts are estimated to represent as much as 10% of aircraft parts on the marketplace (Kotzé and Antonopoulos, 2022). In the summer of 2023, the company AOG 24-part was guilty of 86 instances of forged documents. (Sean Broderick, 2023), including EASA Form 1s for high valued engine components. These engine items with fake documentation were found 96 engines, a number that might increase with further inspections and investigations. The rampant counterfeits issue is not successfully addressed by existing controls.

For part storage facilities, regulations are enforced through audits. Audits of stores are conducted by airline's quality departments on an annual to bi-annual basis according to the responsible national aviation authorities' guidelines. The audits verify that all materials, parts, and components are regularly recorded, and stock checked, with each item tracked through stock management and record keeping software such as the System Applications and Processes in Data Processing (SAP). These audits are conducted by random sampling of parts, tools, and time frames. No continuous data is analyzed, thus giving leeway to unidentified errors. For instance, oxygen bottles and electrostatic sensitive devices (ESD), such as a flight control computer, must be kept within a strict temperature and humidity range. A one-day peak in temperature and humidity levels can render such parts unserviceable, incurring high costs to the airline. However, the chances of the next audit specifically enquiring about climate conditions on the exact day the error occurred are sparse. In addition, only grave findings, labeled level 1 findings, such as parts out of calibration, must be reported by the airline's quality department to the overseeing national aviation regulatory body. With this audit structure, it is left to the airline's quality inspector conducting the audit to decide whether a fault qualifies as level 1, a decision often biased by their personal relation to the manager in charge of the engineering station, both co-workers at the same company. Finding

faulty parts when attempting to install them in a quick turnover is detrimental to the airline's revenue. However, and most importantly, the fault in the parts caused by improper storage might not be detected until the aircraft is in the air, endangering lives.

In aviation, parts are often fixed and recirculated, either within a company or through part vendors. In fact, the lifecycle of a part often involves multiple companies and storage facilities around the world. The only common practices and window into the conditions in which parts were kept are regulations and forms, respectively, mandated by the homogeneous guidelines of national aviation regulatory bodies around the world and audited as previously described. When it comes to reporting flaws and finding innovative solutions, companies rarely emphasize stores. The warehouses, although at the technical heart of the operation, in most cases are not optimized for efficiency, do not receive investment in qualified stores keeping personnel or modern automated tools. Investments in engineering center on retaining personnel and training teams to obtain necessary type ratings, a challenge, especially in a post-COVID industry. Some limiting factors do exist, besides audits, an aircraft with missing tech logs for instance, sees its resale value diminished. But these far-down-the-line taxes are viewed as minor issues by airlines. Furthermore, if the consequences of storage and documentation errors will reveal themselves immediately within the causing airline's operation or further along the line, at the next owner's, is unpredictable. When thinking beyond one airline and considering the life cycle of parts, the problem is exponentiated, opportunities for undetected error in storage conditions and mistakes in paperwork abide. Thus, the true consequences and cost of poorly stored parts and lenient quality checks are difficult to trace as they ripple through the whole aviation industry.

1.2 Hypothesis statement and Study Objectives

There is a problem in traceability and accountability in the aviation supply chain. Storage facilities in commercial aviation's engineering departments, a crucial link, lack efficient inventory management systems. Here, we hypothesize that implementing a blockchain framework for inventory management in an airline's engineering operation increases accuracy of part's locations and facilitates maintenance tracking, promoting cost-control.

The study will be conducted at British Airways Engineering operation at JFK Airport, which contracts local maintenance and is responsible of inventory holding for six other international airlines. The study's objectives are to:

- understand the current strengths and limitations of spares management at BA's JFK station, which serves BA Engineering and its client airlines,
- build a user-friendly inventory management system, capable of tokenizing spares,
- test the viability of tokenizing spare parts in BA's hangars,
- qualify and quantify the impact of the proposed solution to tokenize spare parts.

2 Literature Review

An inventory management system is defined as a group of rules and controls that detect, manage, and maintain inventory level (Jacobs and Chase, 2018). Because this study's objective is to build a stores-keeping solution that serves the needs of British Airways Engineering operation at JFK Airport, the review will focus on current inventory management solutions commonly implemented in aviation, novel solutions from other industries with similar constraints and governing regulations within the civil aviation authority. The aim of the review is to extract recommendations for the study's next steps.

2.1 Current solutions commonly implemented for inventory management.

2.1.1 Current Inventory Management Software: AMOS, SAP, IFS (Maintenix)

In the past 20 years numerous airlines have switched from a patchwork of inventory management systems to company-wide use of one software. British Airways has adopted the SAP system, known as the Engineering Wide System (EWS), a comprehensive and integrated enterprise resource planning solution that streamlines aircraft maintenance across 26 hangars and 142 airports worldwide. Replacing the 150 legacy systems required a substantial effort, involving 23,000 man-days of classroom training, on-the-job guidance from 1,000 system advisers, and 200 process transaction manuals. Despite its initial challenges, it ultimately resulted in significant cost-saving and increase in efficiencies (McCue, 2006). Similarly, China Airlines faced challenges with legacy mainframe systems, which were siloed and inefficient. They adopted IFS's (Maintenix), another integrated maintenance software, leading to a 65% reduction in multi-crew job card waiting time in heavy maintenance, 10% increase in line management and maintenance process efficiencies, and 14 additional revenue days per year due to reductions in scheduled and unscheduled maintenance (IFS, 2023). In the past, making the change from legacy inventory management systems to newer, more digitally enabled solutions has benefited airlines.

2.1.2 Inventory applications on mobile devices.

AMOS, a Swiss Aviation Software spin-off of the defunct SwissAir, prides itself of representing "Swiss Quality at its best" and provides software solutions for Maintenance Repair Overall (MRO) to a large and diverse clientele of 245 airlines and independent MRO providers of all sizes across the world. AMOS has recently introduced a customizable mobile version, see Figure 1, for inventory management; however, it does not run on the blockchain. The company highlights the benefits of the app as producing a "Decrease of loan costs and AOG orders due to wrong inventory, reduce likelihood of delays" (AMOS, 2024). This report shows a trend and a demand for practical mobile applications.



Figure 1 - AMOS Mobile Application, reproduced from AMOS, 2024.

2.1.3 Automated custom mobile apps for inventory management in other industries

In other industries, custom mobile applications for store keeping are commonplace. Ubald et al., 2022, documented their development process for a mobile app prototype for addressing inventory management challenges in the pharmaceutical industry, applying the RUP development methodology and testing various computer tools. The app prototype was built after a detailed analysis of the functionalities needed, to fulfill usability requirements while maintaining a high level of security in a well-presented format. The prototype was then evaluated via a survey of the users and experts. Their results included 92% of users surveyed stating that they completely agreed with the authors regarding the prototype's high security, usability, and functionality. Although the phrasing of the questions could be quite leading, overall customers and experts alike appeared satisfied with the prototype. Interestingly, app aesthetics played an important role in user satisfaction. The systematic approach in development and testing used by the authors is of interest when developing inventory management applications, especially in domains that require high security (Alfredo Leonidas Vasquez Ubald, 2022).

In turn, Yankah et al., 2024, reviewed 38 mobile applications for document management in construction projects. The authors further narrowed their focus to the 21 applications among these that demonstrated best relevance, user friendliness and compliance with the app's goal. In doing so, the authors explore challenges faced by the construction industry's current Document Management Systems (DMS). The paper recommends an ideal mobile app solution capable of streamlining workflows, incentivizing collaborations, and providing access to documents in real-time, these 3 core characteristics ultimately enhancing accuracy and efficiency. Their recommendations can be generalized to solutions for inventory management in other industries where documentation is of essence (Yankah et al., 2024).

In line with developing digital transformation tools, QR codes can streamline the process by instantly updating and accessing item records in a central database, simplifying tasks like adding inventory, tracking movement, and conducting audits. Morales-Hernandez, 2021, provides a step-by-step guide on setting up a QR code inventory system, emphasizing benefits like quick data entry, real-time updates, and cost savings by using smartphones instead of specialized scanning equipment. This approach is interesting for any inventory management by highlighting the role of QR technology in enhancing operational efficiency and accuracy (Morales-Hernández et al., 2021). However, efficient QR coding for managing inventory also implies methodically organizing physical stores, a domain in which research has also delved, notably in respect to the Dewey-Decimal System (Satija, 2013), based on subject matter and widely employed in libraries, or the Hierarchical Location System.

2.2 Blockchain technology for inventory management

2.2.1 Blockchain – leveraging a centralized computer application with the decentralized system of blockchain.

Decentralized applications, making use of blockchain technology, have also been considered for inventory management purposes across industries.

The term Blockchain was popularized by Satoshi Nakamoto from his whitepaper “Bitcoin: A Peer-to-Peer Electronic Cash System”. This was the premise for a decentralized digital currency. The paper highlighted problems with traditional online payments and the need to overcome a trusted third party when transacting. It highlights a system called proof of work, where the validity of transactions and the creation of new Bitcoins are ensured by miners performing computational work. These miners are in turn rewarded with new bitcoins, incentivizing them to conduct computational work to keep the system going. The system uses the basis of cryptography, by computationally time-stamping digital transactions by hashing them into a continuous chain of hash-based proof-of-work, forming an immutable record that cannot be changed. The proof-of-work which validates the transactions is grouped together into blocks by miners. Independent miners are responsible for processing and verifying the transactions. The processing of the transactions is done by solving complex mathematical problems, this is the proof of work concept. When the mathematical problem is solved, a consensus is made between the miners. As the miners are independent, they create a decentralized network it negates the need for a trusted third party (Nakamoto, 2008).

The decentralized blockchain network run on the Ethereum platform enhances this concept further by allowing prerequisite, smart contracts. Because Ethereum incorporates smart contracts, it is a distributed state machine, as represented in Figure 2 (Pettinari, 2023). The smart contracts are coded using the Solidity programming language (Solidity, 2024). Solidity is based off other commonly used programming languages such as C++, Python and JavaScript. With Solidity you can create smart contracts that transact automatically with the Ethereum blockchain.

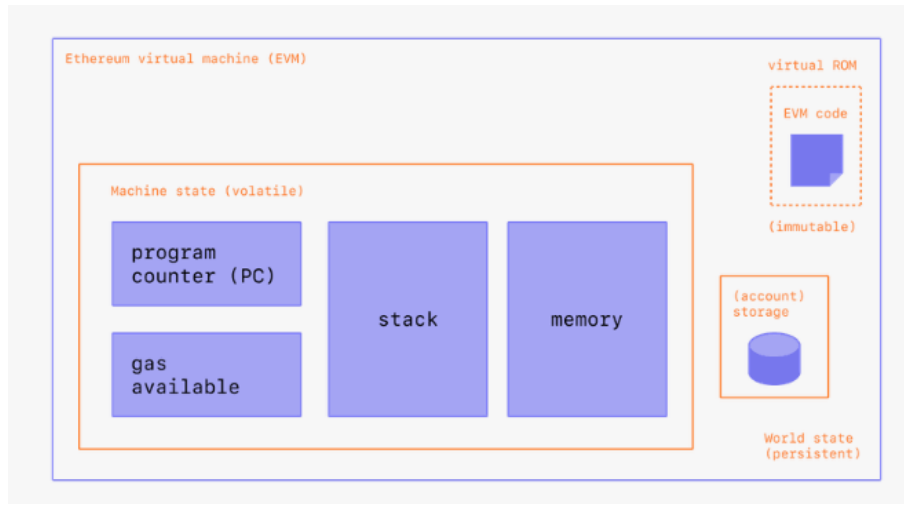


Figure 2 - Diagram of the Ethereum Virtual Machine (EVM), reproduced from Pettinari, 2023.

Ethereum smart contracts are complemented by decentralized applications (dApps), which are programs based on the blockchain. This is fundamentally different to normal applications that run on a centralized system. The dApps run on a peer to peer (P2P) network that is Blockchain based (Cai et al., 2018). Decentralized applications (dApps) offer advantages like fault tolerance, privacy, data integrity, a flexible development platform, and verifiable behavior through smart contracts (Antal et al., 2021). However, they face challenges such as performance overhead, maintenance difficulties, scalability issues, complex user experience, potential centralization, and network congestion (Pop et al., 2021). These pros and cons highlight the balance between the innovative features of dApps and the practical considerations in their implementation and use.

2.2.2 Use cases in industries other than aviation.

Much has been published and speculated on benefits of blockchain technology for inventory management, generalizing across industries. In fact, blockchain technology is believed to have potential to revolutionize inventory management by providing a secure and transparent way to track and manage inventory (Henry and Kathawate, 2023). By leveraging blockchain, it has been claimed that companies can increase transparency, accuracy, and security of inventory management operations (Bhushan et al., 2023).

Some direct use cases of blockchain technology for inventory management have been further detailed. For instance, blockchain can be used to enable environmental, social, and governance (ESG) tracking through supply chain traceability (Henry and Kathawate, 2023). Additionally, blockchain can help companies proactively detect and mitigate supply chain risks before any severe impact occurs.

A paper titled "A System for Storing Anonymous Patient Data in Healthcare applications" introduces a solution aimed at enhancing the privacy and security of healthcare data through a blockchain application (Oksuz, 2024). The significance of anonymizing patient data to ensure secure access and sharing of health records was emphasized. The method of employing pseudo-random identities for patients to mitigate the risk of health record leaks was discussed, highlighting its potential to substantially improve data privacy. This approach was compared with existing data storage practices, illustrating its potential impact on the confidentiality and integrity of patient health information in the digital healthcare landscape. The paper's methodology and proposed solution were critically examined within the context of current trends in healthcare data privacy, underscoring its relevance and possible implications for future research and technological advancements in healthcare information systems. Using the blockchain to encrypt data, adding an extra layer of security would be beneficial to the aviation industry who has struggled with forged parts.

Additionally other industries such as Agriculture have been using the IOT framework integrated with blockchain to enhance transparency and trust in the agricultural food supply chain. A paper called "A Framework for Agricultural Food Supply Chain using Blockchain," discusses how blockchain technology can overcome challenges related to tampering, traceability, and the complexity of global trade, thus ensuring food safety and integrity. The framework aims to leverage blockchain for secure, transparent transactions and IoT for real-time data collection, enabling a comprehensive view from farm to consumer. The paper covers related works, blockchain's benefits and limitations, and outlines a specific proposed framework for applying blockchain in the food supply chain (Sudarssan, 2024). The relevant data being collected is similar as aircraft item batch numbers are mirrored in the industry and storage being an important factor.

2.2.3 Upcoming solutions in the aviation ecosystem incorporating blockchain technology.

Ho et al., 2021, built a system on Hyperledger Fabric and Hyperledger Composer, which is an Ethereum backed system. The system introduced generates a digital twin following the validation and endorsement of transactions. It serves as a practical example for the entire aviation industry, establishing connectivity between Original Equipment Manufacturers (OEMs), airlines, and Maintenance, Repair, and Overhaul (MRO) entities. This integration is designed to support comprehensive 'back to birth' records, streamlining the traceability and management of aircraft parts. By linking these key segments, the system facilitates enhanced coordination and efficiency across the aviation supply chain. The proposed system integrates Radio Frequency Identification (RFID) technology into aircraft parts during their manufacture by OEMs. These integrated RFIDs serve multiple roles, primarily storing essential data about the part, providing essentially a birth certificate. They also facilitate logistics providers in managing inventory and enable airlines and MROs to access and augment traceability records with specific information. A performance evaluation was done to demonstrate the blockchain based system's ability to scale using simulations. The results indicate a highly

predictable performance behavior under varying demand loads. The study mentions several limiting factors including the integration of the Digital Twin, data privacy and security concerns, performance during real world scenarios, interoperability with existing systems as well as compliance and approval across multiple aviation authorities.

In turn, SkyThread is a consortium-driven initiative aiming to improve data flow in aviation transactions by leveraging blockchain (SkyThread, 2022). SkyThread aims to create a Utility Layer that can provide the full account of aircrafts part history and maintenance, supporting over twenty applications to streamline aviation processes. The company is working with regulatory agencies to create a comprehensive industry standard. It anticipates creating revenues of approximately \$11 billion, reducing operating costs by \$6 billion, inventory costs by \$14 billion, and increasing residual asset values by \$2 billion.

The multinational information technology company Society of International Telecommunications in Aviation (SITA), tested a “Flight Chain” project which aimed at using blockchain to facilitate a single, reliable source of truth for flight status data, addressing the issue of multiple and often conflicting data sources in the industry (SITA, 2017). Currently, SITA claims to service about 90% of the world’s airline business and a revenue of \$1.3 billion a year, underscoring the potential impact of this novel solution. For the company, however, the benefits were mitigated by the complexities of the invention. In fact, with every new customer airline and airport, additional nodes would need to be created in a manual and ineffective process, preventing scalability. Since publications of these results in 2017, no further communications on the matter or advances on the project have been released by SITA.

2.3 Cybersecurity risks and regulations within the Civil Aviation Industry

All inventory management solutions and new technologies implemented in aviation must be zealous of potential dangers. In fact, civil aviation has been prone to cyber-attacks. The 5 top recent occurrences include a BA breach of 400,000 records in 2018, a SITA breach of 2 million records in 2021, an EasyJet breach of 9 million records in 2020, a Cathay Pacific Airways of 9.4 million breaches in 2018 and finally an Air Canada breach of 20’000 records in 2018 (cnsight, 2021).

A report from the Civil Aviation Cybersecurity Subcommittee provides further recommendations for improving cybersecurity in the aviation supply chain (Schwindt, 2023). It addresses the complexity and risks associated with the global and interconnected nature of the supply chain, emphasizing the increasing threats from hardware and software vulnerabilities, insufficient supplier vetting, and global access to aviation components. The report outlines objectives to improve supply chain safety, security, and resilience, including developing guidance for cyber supply chain risk management, establishing supplier trust, and implementing secure configuration management. It highlights the need for harmonized regulatory guidance and industry standards. Emphasis is placed on how the aviation industry

organizations must develop and implement effective anti-counterfeit policies and procedures. Counterfeit components include components that have been produced by an unauthorized party and not to specifications, components that are illegitimately sold as a higher specification component (e.g., with a higher environment rating, or components that are not permitted to be resold as they are from an aircraft that has crashed. Brokers may be considered the highest risk of sources of components, especially when these are not subject to aviation requirements for tracing component lots. Among others, the Commonwealth Nations are now required to submit attestation documentation related to protecting against counterfeit components.

Smart technologies can counter the risk of counterfeit components entering the supply chain. Reports of cyber-security incidents across the aviation sector between 2001 and 2021 have been systematically analyzed in a study by Ukwandy and colleagues, 2022. The study finds that the most prominent threats have been from groups that, paired with a state actor, have attempted to gain unauthorized access to information technology infrastructures, with the goal of using the intellectual property thus obtain for their partner nation's advancement. Authors point to the risk of multiplying potential attack surfaces with increasing computer-based IT systems and the practice of Bring Your Own Device (BYOD), valid for both customers and employees. The increased exposure that comes with automation can be mitigated, at least in part, by implementing cyber-hygiene measures, which include recurrently updating of system and anti-virus software, training personnel, and enforcing policies regarding data uploads and passwords. A sense of urgency is necessary as AI technologies develop and with the automated attacks take center stage. (Ukwandu et al., 2022). In developing a smart application for use within the airport environment and on aircraft during maintenance, these questions are essential to avoid creating additional cyber-security vulnerability in an era of relentless and potentially automated threats. As such, literature regarding cyber security risks and management in civil aviation showcases the dichotomy between how automation, IoT technologies and the increase in the use of data can optimize part management and significantly reduce the threat posed by counterfeit components, whilst also potentially providing additional attack surfaces for cyber-attacks, creating new cyber-security risks.

2.4 Conclusion of the Review

Publications regarding mobile applications for inventory management across industries highlight the importance of using a systematic approach in developing and testing prototypes. Key aspects to focus on are streamlining workflows, creating a collaborative interface, and providing real-time accessibility to the data. To provide such accessibility, a QR system approach, coordinated with a reorganization of the stores can be valuable. Decentralized applications appear as a valuable interface allowing for mobile, user-friendly, front ends to have their data secured by a blockchain, as summarized in Figure 3. Recommendations extracted from this literature review are detailed throughout the results section as they guide the project forward.

3 Methodological Approach

The aim of this project is to build, deploy and assess the outcomes of an inventory management solution that interacts with the block-chain and is custom-built for BA's engineering operation at JFK Airport (see 1.2). The approach taken is inspired by the literature review and represented in Figure 4.

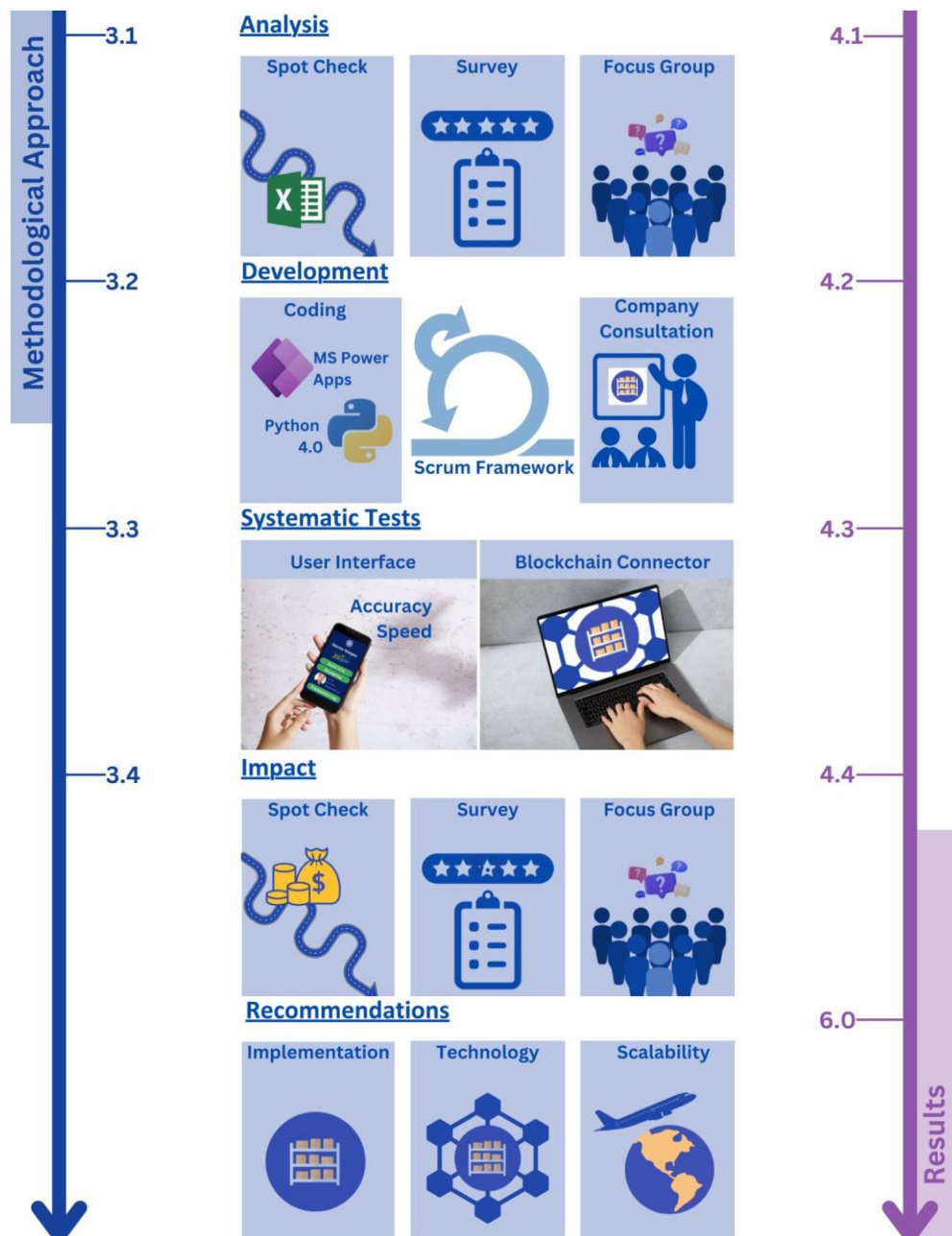


Figure 3 - Methodological Approach Including Analysis of the Stores at Time 0, Development of a Novel Solution, Systematic Testing of the Prototype and Estimation of its Impact, with the Aim of Extracting Useful Recommendations for Future Implementation and Scalability.

3.1 Analyzing the current inventory management spreadsheet system at JFK

Prior to building a custom system, the system initially present was analyzed (Figure 5).

The analysis includes:

1. A detailed description of the system's flows and limitations.
2. A spot check to estimate location errors. The cost of such location errors was extrapolated from tabulated parts pricing.
3. A JFK Stores Inventory Management Feedback Survey deployed to all Engineering team members anonymously.
4. A focus group with all team members and semi-structured interviews with targeted senior unit staff to discuss the results of the survey and delineate a new solution.

The aim of the survey and focus group was two-fold, to engage key stakeholders early on thus fostering a sense of involvement and investment in the project that would favor its later adoption and consulting with key stakeholders from the onset to ensure the project aligned with the organization's goals, addressing real operational needs. The interview sessions aimed at further building trust and open lines of communication to demonstrate the commitment to transparency and collaboration. This is key for a station such as JFK, where recurrent friction arises between unionized and non-unionized teams, working on BA aircraft and customer aircraft, respectively.

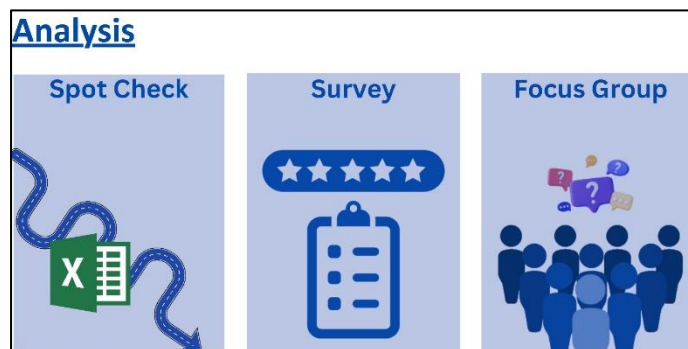


Figure 4 - First Step of the Methodological Approach: Analyzing the Stores at Time 0.

3.2 Developing the solution

A mobile inventory management application was developed using Microsoft Power Apps. In turn, a blockchain component was created with Python 4.0. and its Web3.py library.

The SCRUM framework was employed for app development. With its focus on transparency, communication, and accountability, it can enhance aviation projects by enabling early identification and immediate resolution of safety-related issues through cross-functional collaboration (Rademan, 2023).

Furthermore, consultation with relevant BA personnel in operational maintenance management, in the quality department and in the information technology department was regularly conducted during development. Once the solution was prototyped it was discussed with the Chief Engineering Technology Architect and the Business Improvement and Innovation Manager, directly under the Chief Technology Officer of the airline. See Figure 6.

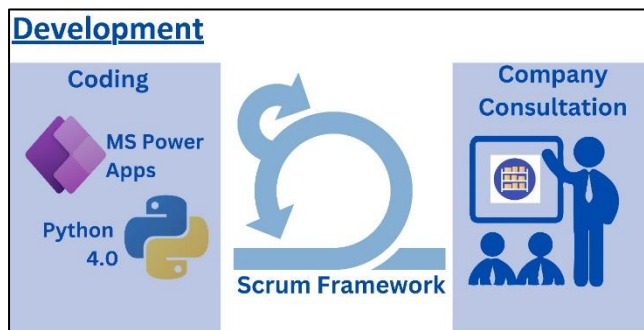


Figure 5 - Second Step of the Methodological Approach: Developing a Stores-Keeping Mobile Application Using Ms Power Apps and a Blockchain Interface, Called Decentralized Application (sApp), using Python 4.0.

3.3 Systematic tests of the App and the Blockchain system

Once a prototype of the mobile app was ready, an experiment was conducted to systematically assess the proposed application. Analysis on the quantitative output of the experiment has as its main outcome (1) the accuracy of the transactions performed and (2) the time taken to complete the transactions. Both outcomes were subject to a repeated measures ANOVA of between-subjects factor role (mechanic v. maintenance reps v. managers), and within subject's factors system employed (app v. analog) and type of transaction (in v. out v. transfer), with a significance threshold of $p < 0.05$.

To test the efficacy of the blockchain twin in preventing inventory errors, a set of simulations, scripted in Python, was conducted. See Figure 7.



Figure 6 - Third Step of the Methodological Approach: Systematic Testing of the Mobile Application and of the Blockchain Interface.

3.4 Evaluating the impact of the application on stores keeping accuracy

A new spot-check, two weeks after deployment of the prototype quantified location errors and their estimated cost. Transfer transactions performed via the app, linked to items being misallocated, were also quantified.

Additional targeted semi-structures interviews with senior unit staff, as performed at the onset of the projected were then repeated. A feedback meeting was ultimately conducted in the same format as the initial focus group, to gather additional information and conclude the project. See Figure 8.

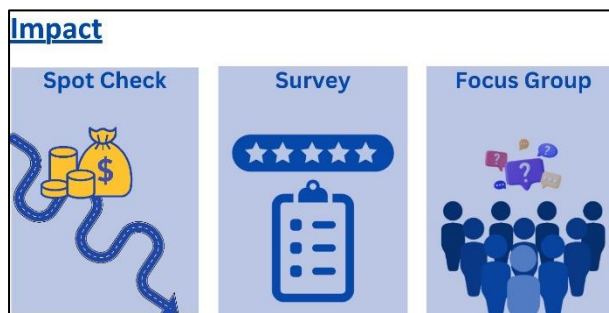


Figure 7 - Fourth Step of the Methodological Approach: Deploying the Prototype and Measuring its Impact with Quantitative and Qualitative Metrics.

4 Results

In this section, we present the findings from a comprehensive evaluation of the newly implemented blockchain-enabled aircraft inventory management system, as compared to the traditional system previously employed at JFK. The objective of the study was to assess the impact of a custom-made blockchain inventory management implementation on various performance metrics, including transaction processing time, accuracy, user satisfaction, and training efficiency among different user roles within the maintenance environment.

4.1 Analysis of the store's keeping system at the onset of the project

4.1.1 Description of the system at the onset of the project

Description of the storage system and inventory management processes are summarized in the diagram below. A full description is available in Annex 1.

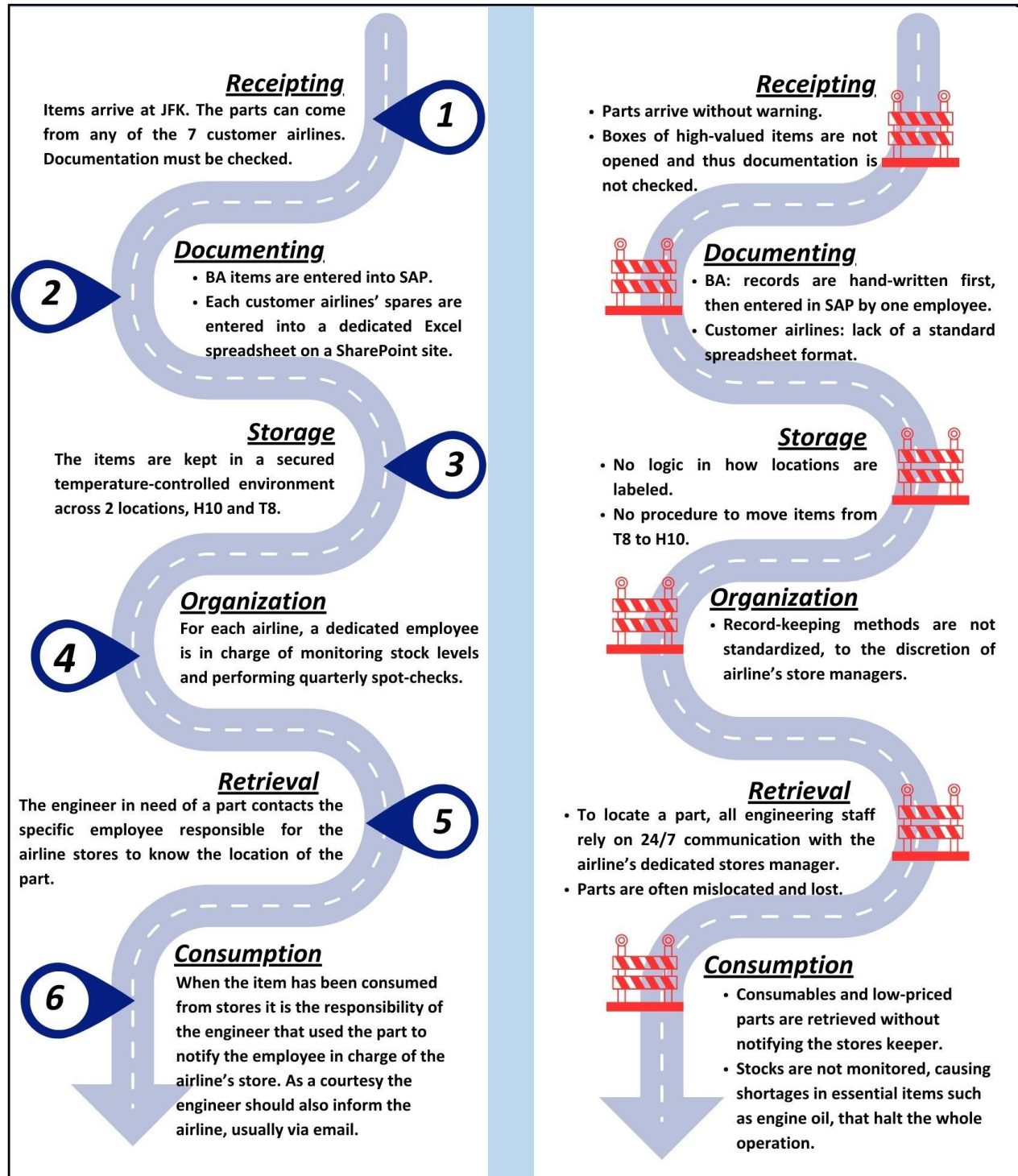


Figure 8 - Diagram of the Information Flows (left panel) and their Main Limitations (right panel) in Stores at Time 0.

4.1.1.1 Strengths and Limitations of the system in place at the onset of the project

For customer airlines, the system affords independence to JFK engineers in charge of each customer airline to best organize their spreadsheets to their liking and understanding, which can favor time-effective completion and tailor to professionals with different digital literacy skills. Furthermore, the record-keeping system is hierarchical, with the local spreadsheet being for internal use at JFK only, while the central record is held by the airline's inventory management system, updated from tech logs and direct reports of maintenance. Such a structure can prevent errors in one record system to impact the other. Lastly, there is no legal obligation as to what specific form of local control over stores must be implemented. As such, spreadsheets on the BA MS platform are compliant with national aviation authority regulations across the world all the while conforming to BA cyber-security standard. For BA's stores, the system is integrated with SAP and thus there is additional transparency for airlines and station engineers. With this, SAP is a system familiar to most engineers since it has been implemented by BA in 2004 and is used for a range of functions.

Several roadblocks can be identified in the current system for customer airlines, summarized above in Figure 9 (right panel). For a complete description see Annex 1.

4.1.2 Staff's views on the system in place at the onset of the project

4.1.2.1 Survey Results and Analysis

The organizational chart of the station is described in Figure 11. The station has 24 employees, of which 19 responded to the survey, or 80%. At least one employee in each level responded to the survey and distribution of responses across levels is representative of the distribution of employees, see Figure 12.

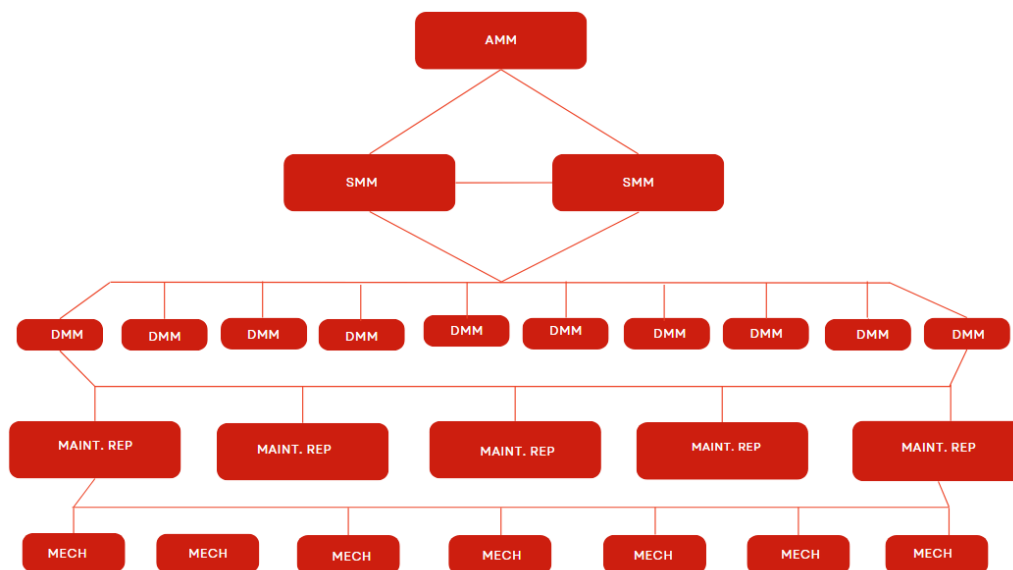


Figure 9 - JFK Engineering Organization Chart.

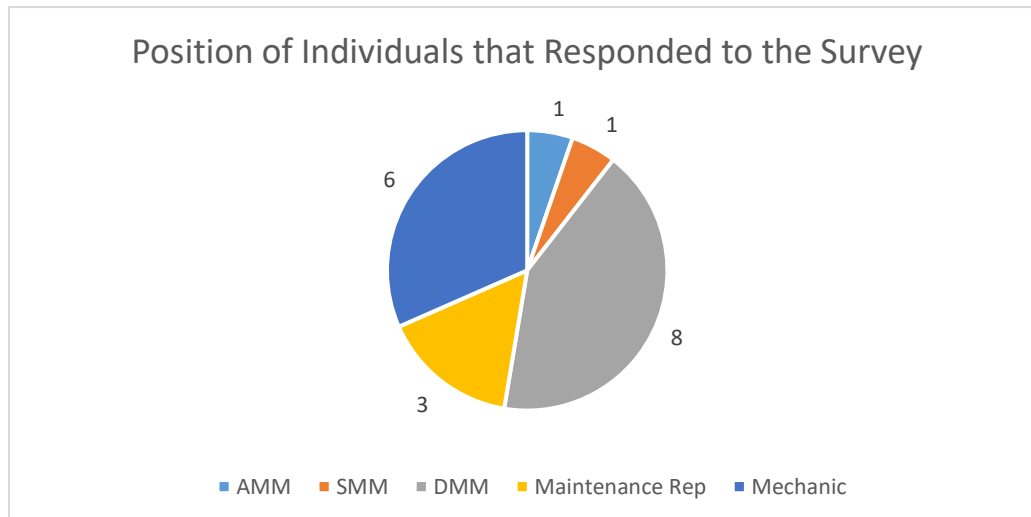


Figure 10 - Distribution of Employees Having Responded to the Survey at Time 0.

The results provide valuable insights , summarized below.

1. Moderate Satisfaction with Current System:
 - The average ratings for questions related to the current state of inventory management (Questions 3 to 18) is 3 out of 5.
2. Challenges in Notification and Documentation:
 - Lower scores on being promptly notified when new items are received (2.53 for Q4) and the ease of locating items (2.89 for Q11) point to challenges in communication and item retrieval processes. The average score of 3 for the accuracy and consistency of documentation (Q5) also suggests room for improvement in data management.
3. General Contentment with Store Maintenance:
 - Higher scores for the physical organization, cleanliness, and layout of the stores (Q8-Q10, Q13) indicate general contentment with the store maintenance and physical arrangement.
4. Operational Efficiency Needs Attention:
 - The efficiency of the check-out process (3.16 for Q17) and experiences of delays or issues in retrieving/checking out items (2.95 for Q18) have average to slightly below-average scores, highlighting areas where operational efficiency could be enhanced.
5. Strong Support for Digital Transformation:
 - Questions related to the implementation of a digital inventory management system (Q19 to Q29) receive high average ratings (above 4), suggesting that employees see significant benefits in a technological upgrade.

It's also worth noting the different roles and their perception between the unionized v. non-unionized managers (DMM/SMM/AMM) – see Figure 13.

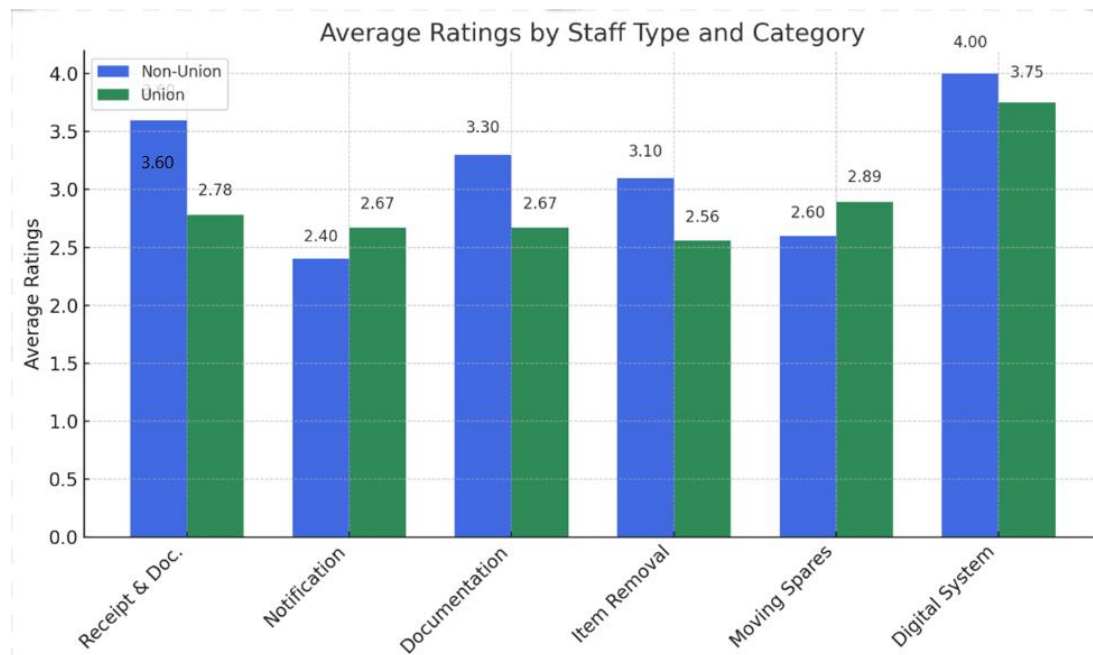


Figure 11 - Bar Graph of Union v. Non-Union Staff Views on the System at Time 0.

In turn, DMM's answered high 100% of the time for Q9,22,23 and 27. This indicates that the DMMs are highly aware of the operational aspects of the inventory system and are supportive of incorporating technological advancements.

In conclusion, these results indicate a recognition of the limitations of the current inventory management system and a strong inclination towards embracing a digital solution that promises enhanced efficiency, accuracy, and convenience.

4.1.2.2 Results of the Focus group

The main takeaway was the team's willingness to digital transformation. A recurring talking point was the importance of reinstating a part-time dedicated stores manager, as previously existed prior to the COVID pandemic, elaborating on how expenditure on part-time minimal wages in exchange for reliable stores-keeping would be a better use of resources than using mechanic's and engineer's time.

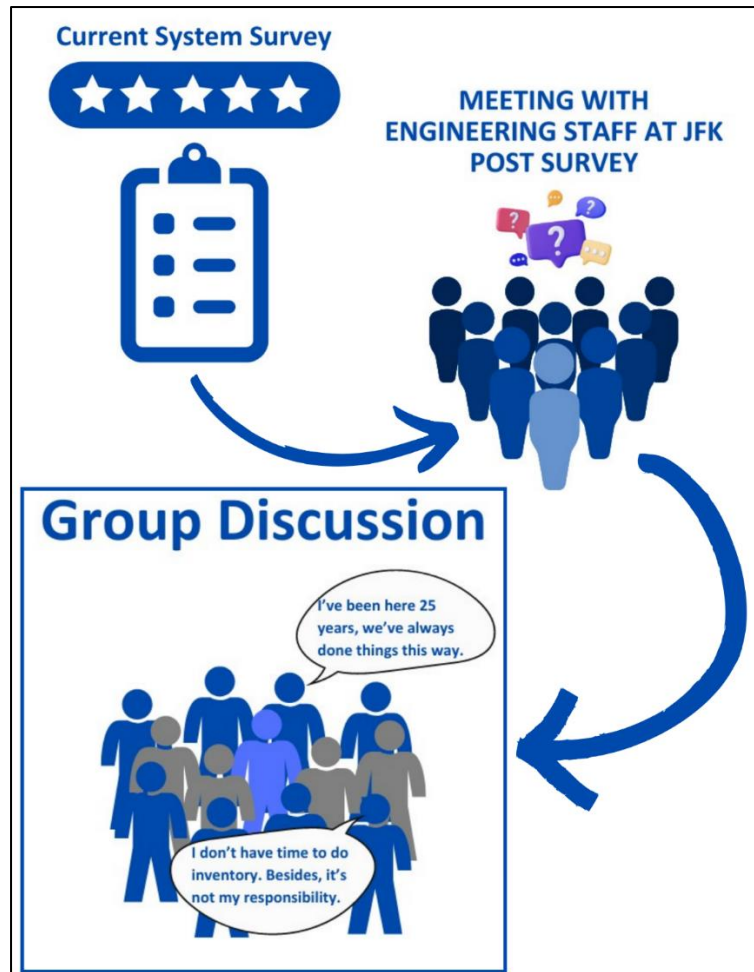


Figure 12 - Illustration of the Qualitative Data Collected Regarding Employee's View on the System at Time 0.

4.1.2.3 Results of targeted Semi-Structured Interviews

Two interviews were conducted, one with the unionized Maintenance Representative in charge of BA stores and the other with a non-unionized Station Maintenance Manager in charge of customer airline services.

The first interview brought to light the burden caused by the current inventory system, expressing frustration regarding misuse of resources, as time should be spent on aircraft maintenance rather than store keeping. When questioned about potential solutions envisioned, the Maintenance Representative cited reinstating a dedicated stores keeper as well as training all employees to complete SAP entries, removing the analog hand-written ledger intermediary and the bottleneck of one employee performing all data entry. In the absence of training for all employees to perform SAP data entries, the Representative mentioned the importance of enforcing compliance to the existing registry system, holding

employees that either failed to complete or inaccurately completed the hand-written book, accountable. The Representative was highly skeptical of a new solution, highlighting the prevalence of a systemic underlying issue, where employees disregarded the importance of storekeeping, independent of the method employed. Furthermore, the Representative questioned the usefulness of a new solution that would not automatically integrate with BA's SAP system, thus diminishing transparency for BA's and failing to address the SAP data entry bottleneck.

The Station Maintenance Manager interviewed had the opposite perspective, showing great expectation as to a potential digital tool. They highlighted the importance of control over essential consumable stocks such as oil and nitrogen, because of instances where the station has run out, having to borrow from other maintenance stations at JFK to continue operating. To overcome this issue, they suggested an automated stock monitoring system, where falling of stock quantity below a certain threshold would trigger an email to the Maintenance Managers in charge, suggesting that a purchase be made. Another issue brought out was the absence of a secure wi-fi internet connection in Hangar 10, rendering use of connected mobile applications difficult. They stated that fiber connectivity would be implemented by February 2024, at the cost of \$50'000, in partnership with American Airlines. Lastly, the Maintenance Manager stated internal budgeting conflicts were delaying the delivery of company tablets to station engineers, forcing many employees to continue using their personal devices on a BYOD fashion. In turn, this has brought about a conflict within the team, since SIM cards and phone plans on personal devices are only paid by the company for non-unionized member with a management role, leaving all other employees to rely on their own phone plans. Awareness of these issues are crucial to building a usable application and to adequately presenting it to team, preempting and appropriately addressing conflictual topics.

4.2 Overhaul of the Stores Organization and Signage

At the incipience of the project, it was observed that the stores did not present a current hierarchical location system, signage was inconsistent and systems distinct between the Hangar 10 and Terminal 8 bonded stores. Storage of items, equipment, and tools at BA JFK stores lacked a standardized approach, leading to inefficiencies in locating and retrieving items. Thus, it was first necessary to overhaul the organization of the physical stores.

A hierarchical locating system at British Airways (BA) Engineering JFK was thus introduced at storage facilities at Hangar 10 (H10) and Terminal 8 (T8). The system defined the location by Store (H10 or T8) > Row (A to Z) > Rack (1 to 10) > Shelf (1 to 100) > Bin (1 to 1000), yielding unique identifiers such as H10 – C – 10 – 1 – 0, see Figure 15.A. Corresponding signage was then created and hung throughout the stores, over all Rows, Racks, Shelves and on Bins, see Figure 15.B. Lastly, a Local Working Procedure (Annex 2) was created and submitted to approval by Quality, being subsequently published in the company's internal documentation repository, called DocuNet, as a mandatory read & sign for all BA Engineering employees at JFK.

Once the stores were reorganized and signs adequately placed, over 350 codes were created for each shelf or bin and attached at their referring location codes were thus printed, laminated, and placed (Figure 15.C)

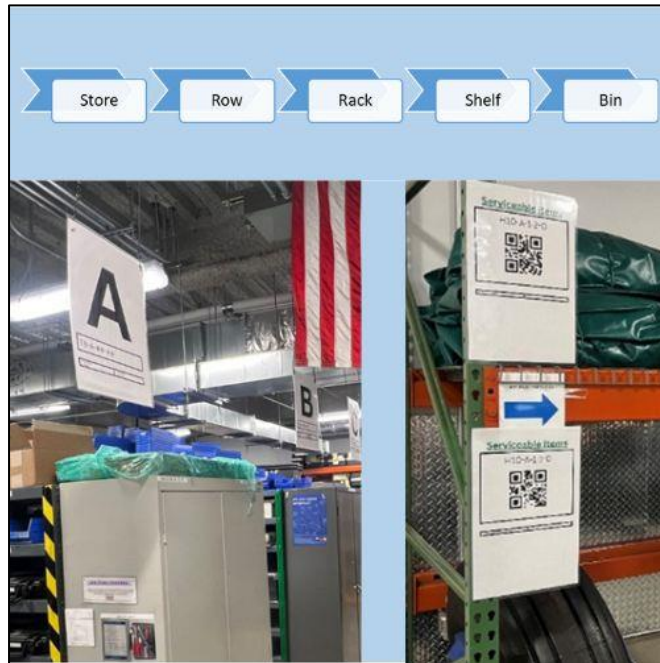


Figure 13 - *Depiction of the Overhaul of the Store's Organization. Top panel: hierarchical location system implemented. Bottom right panel: hierarchical signage throughout stores facilitates navigation and retrieval. Bottom left panel: QR codes on each bin allow for integration with an app.*

4.3 Description of the Mobile Application and Blockchain Implementation

The application harnesses BA's Microsoft (MS) network through MS Power Apps. There is a dedicated BA environment to roll out apps across the company.

4.3.1 Front-end

4.3.1.1 Rationale for the app's front-end

From the literature review and the analysis of the system in place, key messages appeared. Firstly, traceability emerged as a main priority in the front end, allowing all users to access the data before and after each transaction as well as search the database across airlines and fields. In doing so, the front-end would also become a single point of oversight for all users. In addition, traceability could be reinforced by implementation of a QR code signage system within the stores and QR code readers in the front-end. Secondly, increasing accountability was also a priority, transparently stating by which user changes in the databases were made. Thirdly, a user-friendly interface, with an intuitive feeling and a pleasant design, personalized to JFK's station, appeared essential for user satisfaction, potentially boosting adoption.

4.3.1.2 Description of the app's front end

To harness BA's secure MS environment, seamlessly integrate with existing systems and use procedures familiar to the team, the front-end was built using MS Power Apps low-code platform, assisted by its Artificial Intelligence Copilot function.

The structure of the app front-end follows the logic described in the subsequent Figure 16, a user guide (Annex 3), a technical guide (Annex 4) and the full code are also provided (Annex 5).

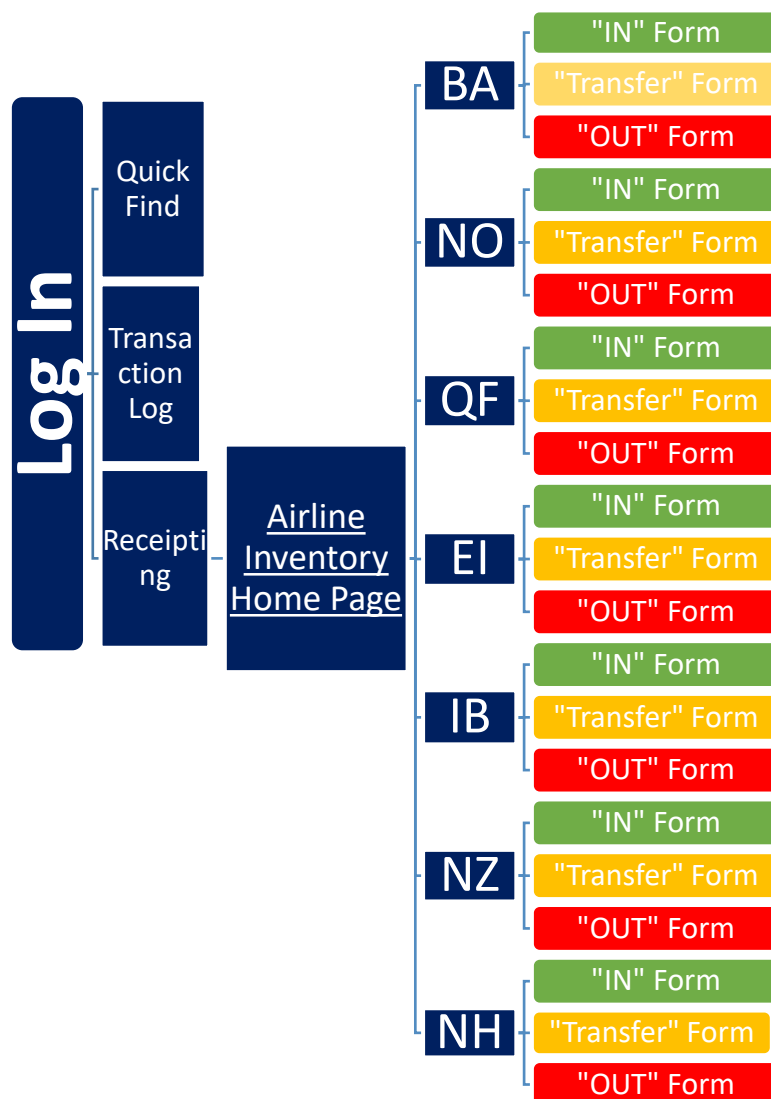


Figure 14 - Diagram of the App Front-End Pages, from Log In to Transaction Forms for Each Airline. Client Airlines Include Norse (NO), Qantas (QF), Aer Lingus (EI). Iberia (IB), Air New Zealand (NZ) and All Nippon Airlines (NH).

4.3.2 Back-end

4.3.2.1 *Rationale for the app's backend*

Research and surveys highlighted the importance of (1) guaranteeing cyber-security, especially in a BYOD context, (2) building upon existing systems to match end-users' digital literacy skills and ease the transition. Using the MS BA environment SharePoint lists emerged as the logical solution, since their functioning is very similar to the Excel Spreadsheets employed by engineers to keep an internal record of each customer's airlines stores. The choice to move from Excel to SharePoint was due to the storage capacity of the latter being much larger and processing faster than with the former (Guzowski, 2019).

4.3.2.2 *Description of the app's backend*

The backend relies on individual airlines' SharePoint lists, that are updated as transactions go through, see Figure 17. The lists are still directly accessible on the MS BA environment, but now in a read-only format for everyone except the engineer in charge of each airline's stores keeping. All columns and formats are harmonized across airlines SharePoint lists. Rules are set so that fields left blank when manually adding an entry to the SharePoint lists are automatically set to N/A or 01/01/2099 when it comes to Expiry Date.

In addition, the back end includes a master SharePoint list of all possible locations for each airline, that is consulted by all Location fields across IN, OUT and TRANSFER pages in the front-end, to ensure a part is not input, output, or transferred to or from the wrong airline's inventory. Importantly, transfers of parts between airlines are not directly allowed by the application since it requires specific authorization from both airlines, unless part of a pool agreement.

Lastly, it also includes a SharePoint list for the transaction log. Any new entry in the transaction log triggers an automated email comprising the entry's content in the body of the message, sent to the researcher. The transaction Log is available to view to all members of the JFK BA team, within the MS BA environment, only the researcher who created it has editing capabilities.

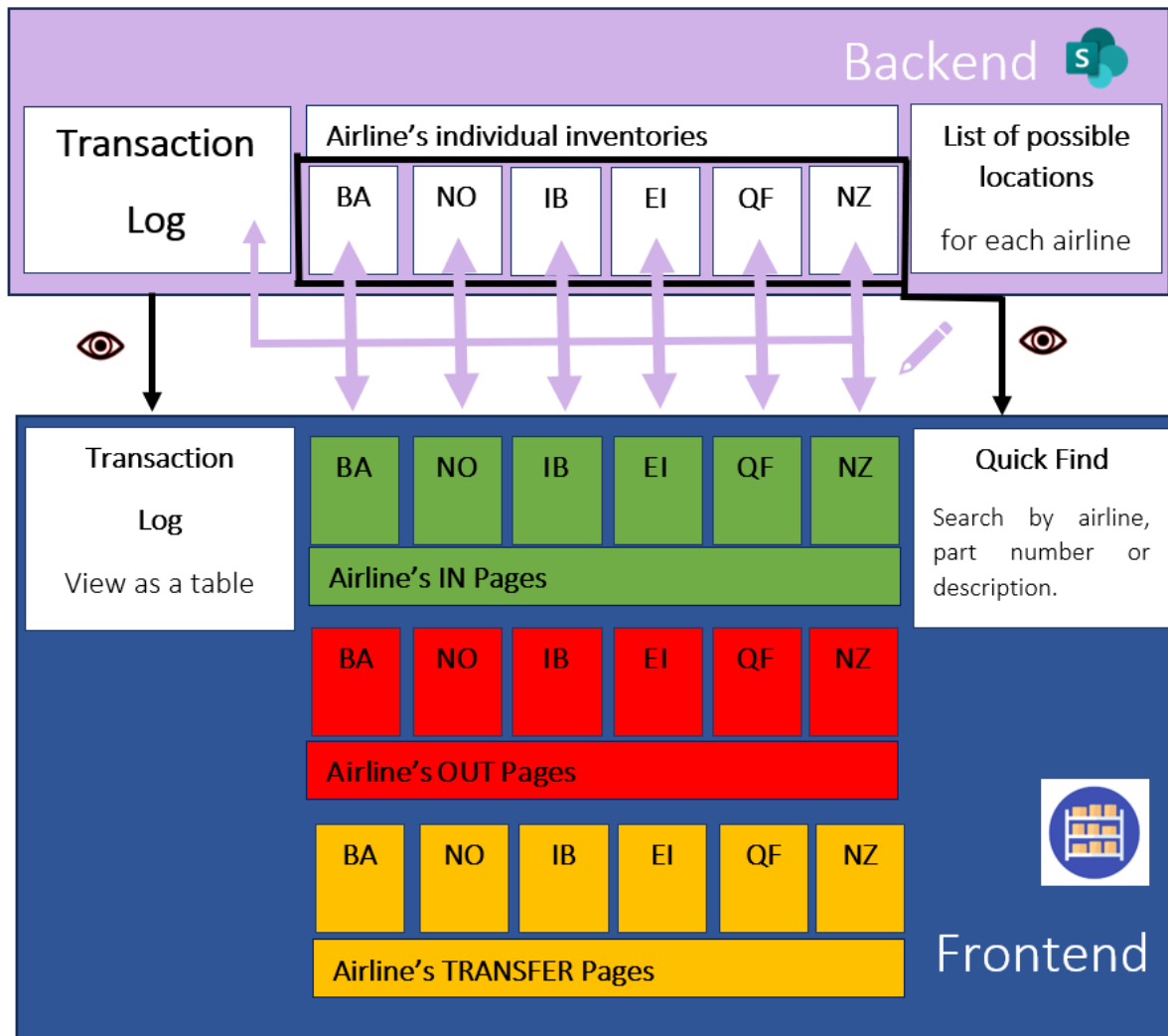


Figure 15 - *Diagram of the Mobile Application's Frontend (Blue Panel) and Backend (Purple Panel). Black arrows indicate view-only information, while purple arrows indicate information can be edited. Airlines are represented by their tail numbers, see figure 16 for a list of abbreviations.*

4.3.1 Interface with the blockchain

4.3.1.1 Rationale behind the blockchain interface

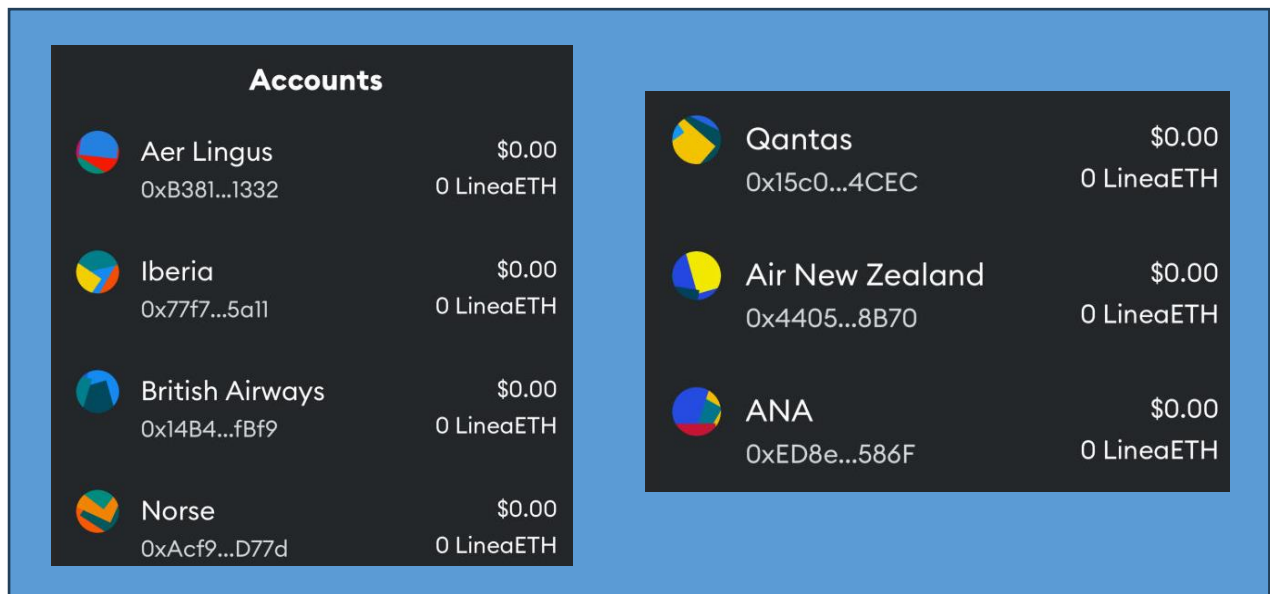
According to the literature search, blockchain is best harnessed for secure inventory management by (1) creating a digital twin of records on a distributed ledger and (2) tokenizing items to be transacted and (3) specifying transaction mechanisms and rules in a carefully crafted smart-contract based on a buyer-seller relationship. In this project, each item was tokenized by associating it with a unique identifier, created by sequentially combining the strings that make each record distinct in the application's back-end databases. Transactions of these tokenized assets are made between each airline's catalog, the distributed-ledger

twin of the airline’s SharePoint list in the application’s back end, and an additional catalog that represents all the application’s users and is the mirror image of the log. A smart contract is deployed, restricting which airlines the universal user is allowed to transact with, which specifications regarding the part are mandatory for a transaction to go through and how quantities and entries in the ledger are updated for each type of transaction.

Because of security concerns, no low-cost option was available to interact with BA’s MS environment. A rest API was envisioned, and a template coded, but its implementation was ultimately limited by paywalls within MS Azure. As such, to allow the prototype to go forward, full integration between the application and the interface with the blockchain was not achieved at the project’s conclusion.

4.3.1.2 *Wallet Creation and Management*

For each airline, including British Airways, as well as for a unique virtual user called Stores Keeper, a unique digital wallet is created on MetaMask, a popular Ethereum wallet platform. Each wallet is secured with a private key, which is critical for authorizing transactions (Figure 18). The security of these keys is paramount and is typically managed by a small group of authorized personnel within each airline's inventory management team.



Accounts		
	Aer Lingus 0xB381...1332	\$0.00 0 LineaETH
	Iberia 0x77f7...5a11	\$0.00 0 LineaETH
	British Airways 0x14B4...fBf9	\$0.00 0 LineaETH
	Norse 0xAcf9...D77d	\$0.00 0 LineaETH
	Qantas 0x15c0...4CEC	\$0.00 0 LineaETH
	Air New Zealand 0x4405...8B70	\$0.00 0 LineaETH
	ANA 0xED8e...586F	\$0.00 0 LineaETH

Figure 16 - MetaMask Wallets for Each Airline.

Users of the mobile application are not attributed to a unique individual digital wallet since such structure would require automated mechanisms for wallet creation. This would engage third party web services and be associated with security risks, potentially becoming a gateway for attacks to BA’s MS environment.

Each blockchain transaction has its twin in the log file, a SharePoint list containing the identity and email of the individual responsible for the transaction. Thus, authorized BA personnel, having access to both the SharePoint log file and the blockchain ledger, can easily trace back from each transaction's unique identifier who initially elicited it via the application front-end. Accountability is ensured without compromising security.

4.3.1.3 Smart Contract Integration

The smart contract's architecture is delineated in the diagram below. The full Solidity code and its explanation are available in Annex 6.

The smart contract is coded in Solidity, tested, compiled, and deployed using Remix IDE, on the Sepolia Testnet under ownership of the store's wallet. It includes functions to authorize a new airline's digital wallets for transactions, to receipt items in and out, as well as to consult items present in the distributed ledger.

4.3.1.4 Inventory Tokenization

Each part in the inventory is represented as a digital token on the blockchain with a unique identifier. These tokens carry as metadata part unique identifier, part number, part description, serial number, batch number, expiry date, location, and quantity.

When a new part is added to the inventory, a corresponding token is created and assigned to the catalog associated with the airline's wallet address. This token can then be tracked across the blockchain, providing a transparent history of its movement and usage.

4.3.1.5 Transaction Process on a Custom Decentralized Application (dApp)

Transactions are run using a proprietary script coded in Python 4.0, using the Web3.py library, constituting a dApp (Annex 7).

A node in the blockchain net is connected to using a http link provided by Infura. The script automates reading airline's SharePoint lists and SharePoint log file line by line and performing the corresponding transactions as defined by the smart contract. For every transaction authorized, the information for the part transacted is retrieved and saved in a list. Likewise, every transaction hash is saved in a separate list. In addition, the script includes timers to prevent delays in transactions to disrupt the sequences of transactions to be performed, which could lead to errors in records, for instance causing an attempt at outing a part before the inputting of the part is complete. This feature is of the essence given transactions can take variable times, depending on the demand placed on the net when the transaction is signed. See Figure 19 for a full diagram.

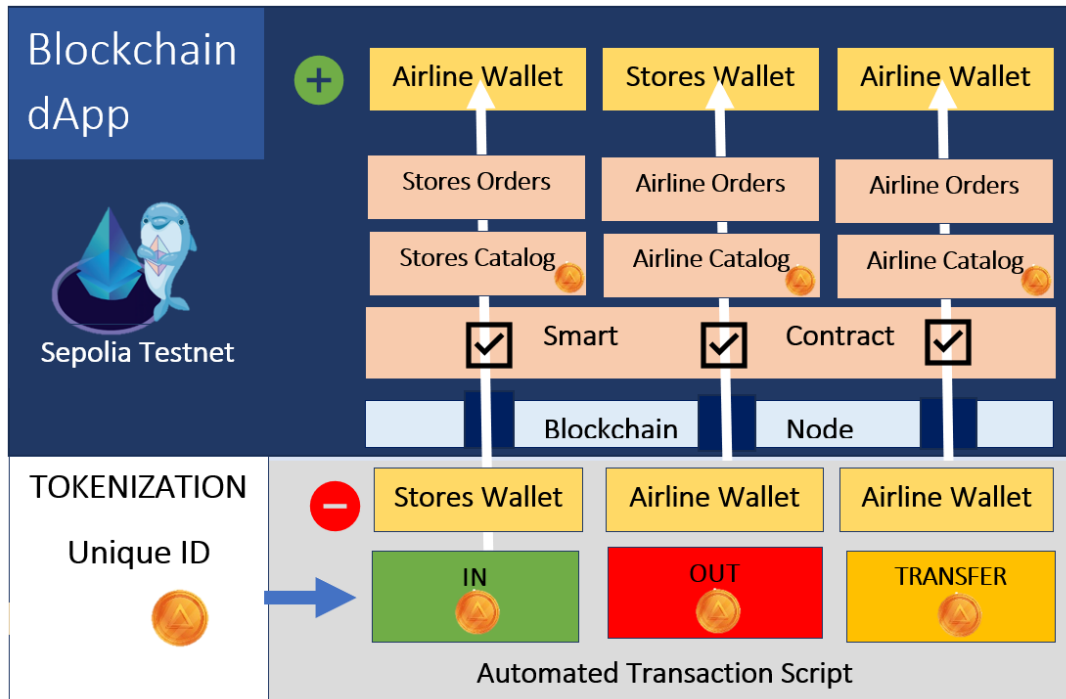


Figure 17 - Diagram of the Decentralized Application.

In the first initiation of the script, or any time a new customer joins the JFK's BA Engineering client roster, the airline's catalogs must be saved in the blockchain. Once the catalogs are set up, transactions can be performed. For this, there is not, at this stage, a direct integration with the mobile application. The log file must be uploaded in the script's environment. For this study, the log file was manually uploaded once at the conclusion of the experiment conducted to verify the usability of the mobile application (see methods 4.5 and results 5.5). The file is then read by the script, and each line triggers a specific transaction, thus updating the airlines' catalogs on the blockchain. For each of these transactions, Ether is debited and credited to the buyer's and seller's wallets, respectively. See Figure 20 for a diagram of the complete solution developed, including mobile application and decentralized application .

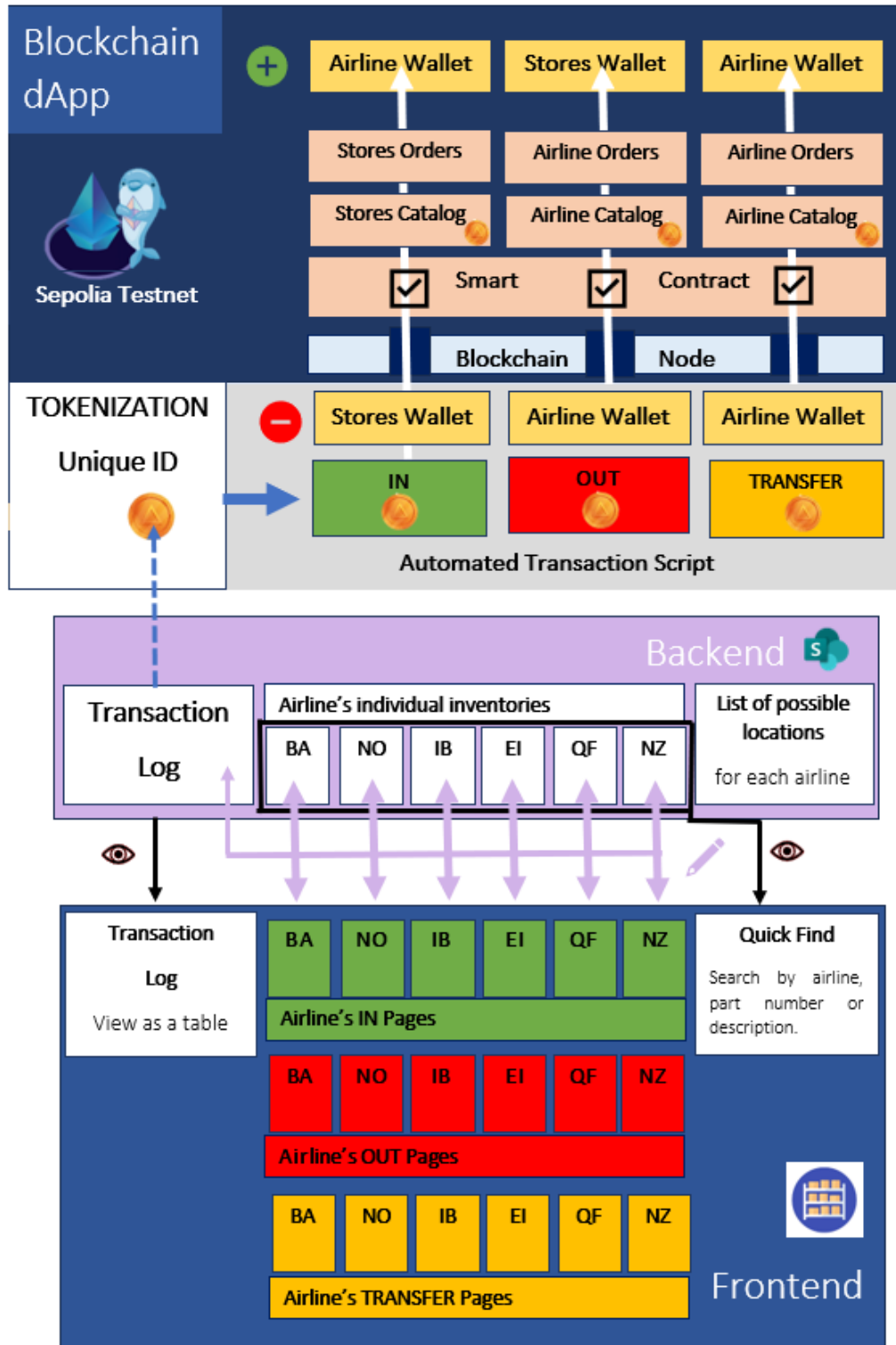


Figure 18 - Diagram of the Stores Keeper Solution. Includes a front end (Stores Keeper Mobile App), a back end (MS Share Point Lists) and a decentralized application coded in Python and running on the blockchain (Sepolia Testnet).

4.4 Evaluating the Usability of the Mobile Application

4.4.1 Materials Created to Introduce the Application to Users

Once the application was ready for deployment, it was introduced to users. A complete user guide was provided (Annex 4). Video demos were recorded (screenshots in Annex 7) and shared on relevant company communication channels to users. A technical guide (Annex 5) was also available for consultation. One-to-one training was additionally available upon request.

4.4.2 Systematic testing of the applications

A structured test of the application was conducted, recording the screens of the mobile devices used. Statistical analysis of anonymized data was performed using IBM SPSS Statistics for Windows.

Eleven (13) users completed the test, performing a total of 18 transactions each. Distribution of subjects is described in the table below, according to relevant categories for the analysis. One subject was rejected from the analysis, due to failure of the screen recording. Full results are available in Annex 8.

Table 1 - Distribution of Employees Having Performed the Experiment to Test the Mobile Application's Prototype.

Category	Mechanics (Unionized)	Maintenance Reps (Unionized)	Managers (DMM/SMM/AMM – Non Union)
N of Subjects	4	1	7

4.4.2.1 Accuracy of Transactions

The primary outcome considered was the accuracy of the completed transactions. The log entries in the back end of the mobile application to verify if transactions were correctly performed either the mobile application, rating them 1 if correct and 0 if incorrect. Similarly, the entries in the analog book were photographed and read by the researcher, giving equivalent rating. When handwriting was difficult to make out, creating ambiguity as to whether entry was correct or wrong, a 0.5 rating was given. Scores were summed for each transaction type, yielding a total of $min = 0$ to $max = 3$ for each of the 4 transaction types.

The mobile application ($m=1.00$, $s.d.=0.00$) was significantly ($F(1, 33)= 12.50$, $p=0.00$, $p. eta sq. =.28$) less prone to errors than the analog system ($m=.85$, $s.d.=.04$). Furthermore, the app was especially beneficial ($F(2, 33)= 4.8$, $p=0.01$, $p. eta sq. =.23$) for mechanics, more so than for maintenance reps and managers (see Figure 21). Interestingly, such divide bridged across the union v. non-union separation of roles. Because only 1 subject was a maintenance rep, we repeated the analysis by adding said subject to the Mechanics group. With this, the interaction between method used and role, confounded with the effect of union status,

became marginal ($F(1, 34) = 3.43$, $p = 0.07$, $p. \eta^2 = .09$). No other within-subject's factors, between-subjects factors or interactions reached significance in either analysis (all p 's > 0.4).

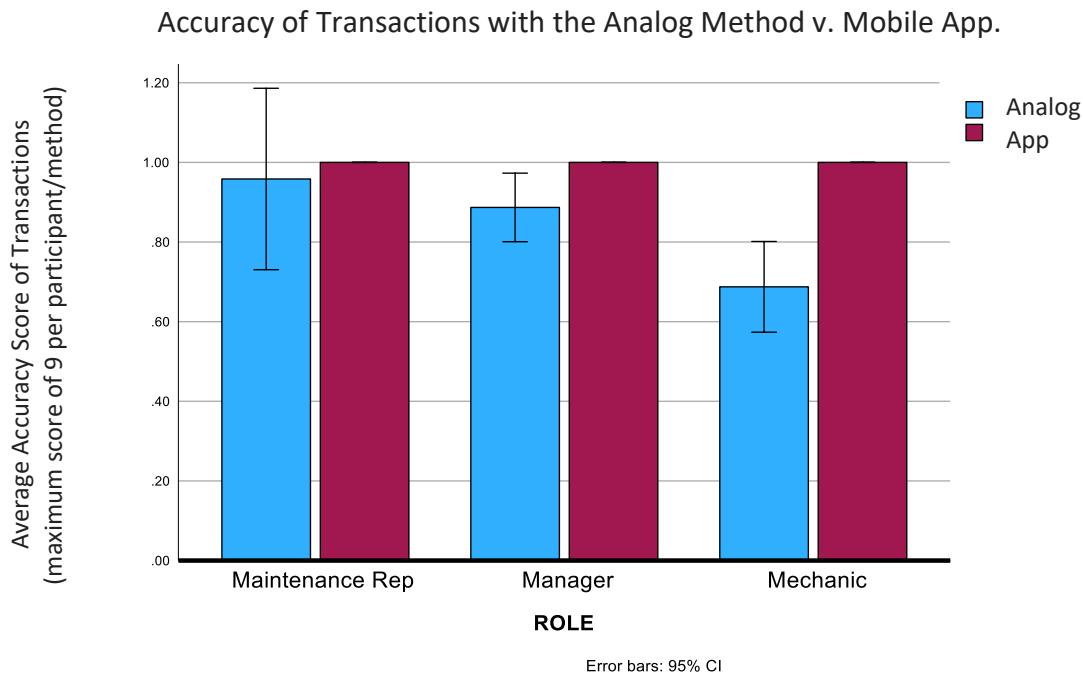


Figure 20 - Accuracy Score of Transactions by Employee Category for the Analog Method as Used at Time 0 (Blue) and the Mobile Application's Prototype (Magenta).

4.4.2.2 Time taken to Complete Transactions

The secondary outcome considered was the time taken to complete each transaction. Screen recordings were analyzed by the researcher to extract the duration of the transactions, in both the app and the analog system. Here as well, the within-subjects factor of analog ($m = 64.95$, $s.d. = 2.24$) v. mobile app ($m = 43.92$, $s.d. = 1.17$) reached significance ($F(33, 1) = 75.30$, $p = 0.00$, $p. \eta^2 = 0.70$). There was an interaction with analog v. mobile app ($F(33, 1) = 17.08$, $p = 0.00$, $p. \eta^2 = 0.34$), showing that the app increased the speed of transactions for "IN", "OUT" and "Quick Find" (all p 's $= 0.00$), but not for "Transfer" ($p = .32$). In addition, a main effect of transaction type appeared ($F(33, 1) = 52.48$, $p = 0.00$, $p. \eta^2 = 0.61$), with Quick Find transactions being significantly faster ($p = 0.00$) than the 3 others (by $m = 17$ ms - 24 ms, $s.d. = 1.58 - 2.30$). No other within or between factors reached significance (all p 's > 0.53). This pattern of results attests to a time-saving capability of the app across transaction types and employees, with an increased benefit for certain types of stores transactions.

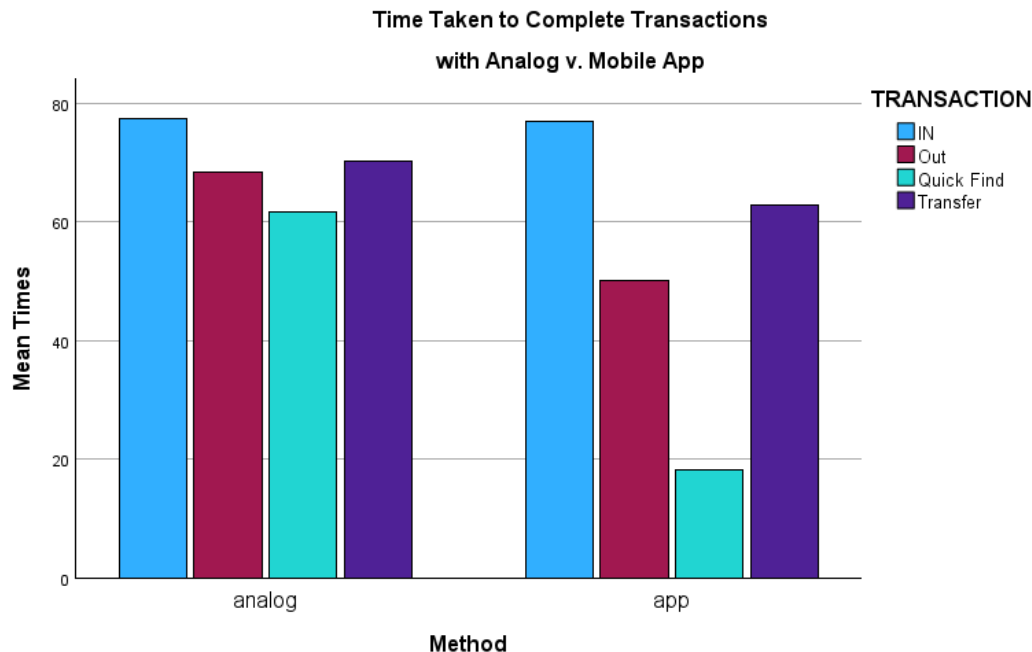


Figure 21 - Box Plot of the Mean Time Taken to Complete a Transaction by Transaction Type, Using the Analog System as Implemented at Time 0 or the Mobile Application Developed.

4.4.2.1 Observed Difficulties Using the Mobile Application

The tertiary outcome considered were patterns of difficulties using the mobile application. Screen recordings were viewed and subjectively analyzed by the researcher. Some common issues were internet connectivity problems, failure to understand the logic of the quick find page and the part location not being typable, having to be input only by scanning a QR code. Of note, these difficulties did not result in terminal errors since all participants completed the 18 transactions required without assistance. See Annex 8 for screenshots of the experiment and a table of results.

4.4.3 Systematic testing of the Blockchain component

4.4.3.1 Simulations

The full simulation script is provided in Annex 9.

For all simulations tested, the blockchain component to transact on the blockchain scored 1, reaching a perfect score of 5. In other words, the blockchain implementation was successful in identifying errors and limiting transactions accordingly for the 3 scenarios listed below:

- Retrieving a wrongly identified part from storage.
- Retrieving a part in a stated location that does not match its physical location.
- Creating a duplicate record when parts with identical identifiers are added to stores, as opposed to increasing quantity of said part in stores.

A Tamper-proof twin of records was thus maintained in the testnet's distributed ledger. However, more rules were implemented in the mobile application's front end than in the Smart Contract, the latter being thus more permissive. This is because both methods were designed to be used in conjunction, the smart contract transacting only from the SharePoint log, as updated by transactions on the mobile application's frontend. A balance was achieved between the smart contract functionality and its complexity.

4.4.3.2 Daily Transactions from the Log to the Blockchain Component

In addition, for the 3 weeks that the application was deployed, new entries in the log were input to the blockchain transacting script once. Because there were 3 or less transactions per day in the log, this frequency was sufficient for the blockchain to detect potential errors. However, no errors occurred in the test period. The average cost of the transactions was \$ 0.03 and the time taken to complete each transaction varying from *min* = 0.4 seconds to *max* = 45 seconds, depending on the testnet's demands at the time of each transaction.

It is of note, however, that the blockchain implementation was not fully integrated with the mobile application, requiring the log SharePoint list to be downloaded and placed in the blockchain component's environment. Likewise, there were no automated mechanisms to notify users of the mobile application when transactions on the blockchain component failed.

4.4.4 Evaluating the impact of the new solution

4.4.4.1 Qualitative Survey of the Experiment's Subject and Focus Group

After having completed the experiment, all 20 subjects filled a survey comprising 5 items. To the question "Did you find what you were looking for?", 100% of responses were "Yes". The transaction experience, an open question received responses that ranged from "Great" to "Excellent". All subjects responded "Yes" to "Did it Save you Time?" and all ratings, from 0 to 10, were 10. No suggestions for improvement were received.

However, of note, this survey was taken in presence of the researcher and with users logged in to the mobile app, potentially biasing answers. Thus, a different and anonymous survey was conducted via MS Forms at the conclusion of the project. Responses collected were $n=19$. Overall app rating on a scale of 0 to 10 painted a mitigated picture ($min=7$, $max=10$, $mean=8$, $s.d.=0.25$). Responders did not consistently perceive the app as capable of preventing parts from being misplaced, only 70% responded positively. Furthermore, only 65% of responders agreed that the application saved time when transacting parts. All responders cited Quick Find as being the most useful feature of the application. Interestingly, only 50% of responders mentioned having used the application on their own at least once. For these responders only ratings were higher than in all responses ($min=9$, $max=10$, $mean=10$, $s.d.=0.5$), and 100% viewed the app as being effective to both prevent errors and save time. In turn, for the same 50% of responses, the most useful feature was, as for the larger cohort, the Quick Find.

4.4.4.2 Spot-Check Results

A first spot-check before the onset of this project (time 0) found 40 parts mislocated and additionally 100 serviceable parts of which the location was absent in records. The value of these parts ranged from $min=\$50$ to $max=\$10'000.00$, with an average of $m=\$1'000.00$ and $s.d.=\$750$, with 75% of the misplaced parts having a value between $\$1'000$ to $\$2'000$. The total potential losses estimated were $\$200'500$.

A spot check conducted 3 weeks after the mobile application was deployed (Time 1), across BA stores and all other customer airline stores found 20 parts mislocated, an 85% reduction in parts misplaced when compared to Time 0. In addition, revising of all SharePoint lists did not reveal any entries timestamped at after the deployment of the mobile application and conducted via the mobile application, since said transactions were also present in the log SharePoint list, that had errors in their entries in either format or content. The value of these parts ranged from $min=\$50$ to $max=\$5'000.00$, with an average of $m=\$1'000.00$ and $s.d.=\$25$, with 75% of the misplaced parts having a value between $\$1'000$ to $\$2'000$. The total potential losses estimated were $\$48'000$, representing a %76 reduction from T0 to T1. See a breakdown by part value in Figure 21.

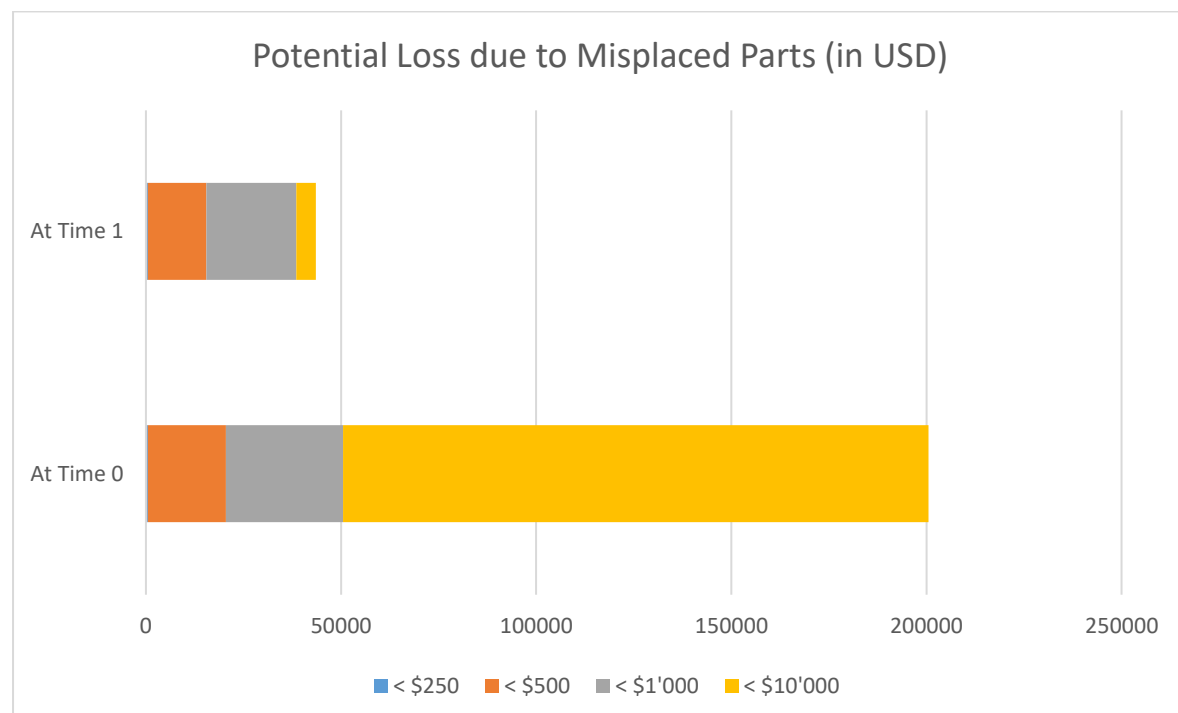


Figure 22 - Bar Graph of Potential Losses Due to Misplaced Parts at Time 0 and Time 1. Summed costs attributed to misplaced parts are color-coded by part value.

4.4.4.3 Markers of Spontaneous Adoption by Personnel

User acquisition was determined from entries in the log that did not stem from the experiment conducted (detailed in methods 4.4) during the first 3 weeks of deployment. In total 3 engineer users were acquired, each conducting one or more “IN” transactions for the airline’s inventory that was under their responsibility. Of note, all the users acquired started using the application spontaneously after taking part in the experiment (see methods 4.4 and results 5.3).

In contrast, by analyzing the time stamps of modifications made on the SharePoint lists in comparison to the timestamps in the Log, modifications that were made directly on the back end, as opposed to through the app’s front-end, were detected. In total, during the 3 weeks of deployment 2 duty maintenance managers systematically used the back end, rather than the front-end, to modify inventory records. These 2 duty maintenance managers were not part of the ones that adopted the mobile application.

4.4.4.4 Feedback from a Conclusion Meeting

Lastly, a focus group was held at the project’s conclusion, as an online Teams meeting, with 15 employees in attendance. Overall feedback over the whole project was positive, from the stores location’s reorganization and relabeling to the adoption of the mobile application as well as the channel of communications opened, and the conversations sparked. Subjects brought out the importance of keeping such communication channels open, especially between union and non-union members. When presented with the results regarding prevention of errors, time saved and potential financial losses prevented, the group expressed what was best summarized as being “shocked but not surprised”. Such group sentiment was different from the average individual responses to the conclusion survey (see results 5.3.1.2) where only 65%-70% agreed with these statements. A presentation of the blockchain transaction portion of the project, never mentioned before, did not trigger any further comments or suggestions. It was further discussed how few users had adopted the tool since its deployment despite its advantages and how it should be formalized as a mandatory working procedure.

5 Discussion

Our findings are discussed in the context of their significance to our research questions, providing a clear picture of the blockchain-based system's effectiveness in revolutionizing inventory management processes.

The project's primary aim was to tokenize spares in a distributed ledger to improve security and efficiency. To achieve this, it had to first overhaul 2 stores physical organization, place over 300 signs, homogenize 7 inventory databases and create a complex mobile application, all the while keeping all stakeholders satisfied. Only after these steps were complete, was centralized and coherent data available for a blockchain implementation.

Reorganizing the stores with hierarchical unique identifiers, presented in text and QR codes decreased time to retrieve and locate items (Morales-Hernández et al., 2021). The mobile application developed, called Stores Keeper, is a versatile app that can easily scale to fit any airline's needs. A similar app has been recently made available by AMOS , 2024, however it cannot be personalized and does comprise a blockchain component. Extensive testing in quantitative and qualitative terms, using methods similar to Ubald et al., 2022 and Yankah et al., 2024, demonstrated Stores Keeper accuracy-boosting, timesaving, and cost-saving potential, even only after 3 weeks of deployment.

The Stores Keeper blockchain component, albeit not fully integrated with the mobile application, was successful at detecting errors in stores storekeeping during simulations. In addition, it proved to be effective in the 3 weeks of its deployment, creating and updating daily an immutable twin of the Stores Keeper back end. The existence of such an immutable record adds a layer of security that physical books, spreadsheets in virtual clouds or SAP-like software's cannot provide (Henry and Kathawate, 2023; Oksuz, 2024; Sudarssan, 2024). In an industry with high employee turnover, low investment in stores management and yet very high safety stakes and costs associated with spares, achieving the utmost security standards in storekeeping is crucial (FAA, 2019).

Furthermore, the blockchain portion of the project, as it was conceived, overcame limitations described in literature. Firstly, one of the lessons of the FlightChain project were that scaling such a network to an ever-growing number of participants , as new airlines were onboarded required extensive work from developers and thus rendered the whole project difficult to justify (SITA, 2017). Here, thanks to the smart contract and the blockchain component created, little time or technical knowledge is required when adding a new airline to the roster, other than setting up a MetaMask wallet and providing the airline's inventory spreadsheet to the blockchain component. Secondly, because transactions through the blockchain and through the Stores Keeper app were not synced in time, delays caused by high demands on the blockchain net did not affect users of the Stores Keeper app, an essential contention point in an industry where time is of the essence (see limitations in (Ho et al., 2021)). This dissociation between the timelines of the Stores Keeper and the blockchain script was not problematic for accuracy of inventory keeping since rules in the smart contract were redundant with those in the Stores Keeper front end. Furthermore, since most rules were implemented in the Stores

Keeper front end the complexity of the smart contract was alleviated, avoiding errors. Thirdly, the solutions was cost-free, as it was deployed on a testnet, which is a blockchain designed to mimic the operating environment of a “mainnet” but is cost-less since it runs on a completely separated ledger (Axen, 2022). Beyond the current prototype stage, deployment in the Ethereum “mainnet” would ensure the record’s perennation, with costs remaining under \$10 / year at the current rate of use (Tambe, 2024).

Importantly, because the Stores-Keeper application was fully created and hosted within the company’s pre-existing cloud environment, cyber-security automatically conformed to company standards, as confirmed by consultations with relevant departments. However, because the deployment of a mobile application enticed a BYOD practice, it is recommended that cyber hygiene training and communications to raise awareness are put in place as the solution is rolled out (Ukwandu et al., 2022). Regarding the blockchain transacting portion of the solution, information stored in the blockchain is made secret through hashes, a form of cryptography (Peng, 2021). In this implementation, a security vulnerability exists in the blockchain component, in the manual step of uploading data from the company’s cloud environment to the transaction script.

To address such vulnerability, it is ideally envisioned that the blockchain component (D’Oro et al., 2022) is moved to the airline’s cloud environment and integrated with the mobile application, via a restAPI (Massé and Massé, 2012) and custom connector (Jopanchal, 2023). Because such system would require additional development cost, for instance with a paid subscription to a platform such as MS Azure (Divyaswarnkar, 2022), prototyping to thoroughly assess the efficacy and the real benefits of tokenizing spares on the blockchain is essential before incurring additional costs. In addition to integrating blockchain and mobile application, an automated parallel path can be imagined for the integration of the mobile application and the SAP software used throughout the company. In fact, according to a consultation with the company’s IT department, when the most recent SAP system update is completed, third-party integration will be allowed, and SAP GUI scripting (SAP, 2019) could be used for the Stores Keeper application to transact directly with the BA SAP system. Here, as for direct integration with a blockchain system, thorough proofing of the prototype is recommended before establishing data-altering automation.

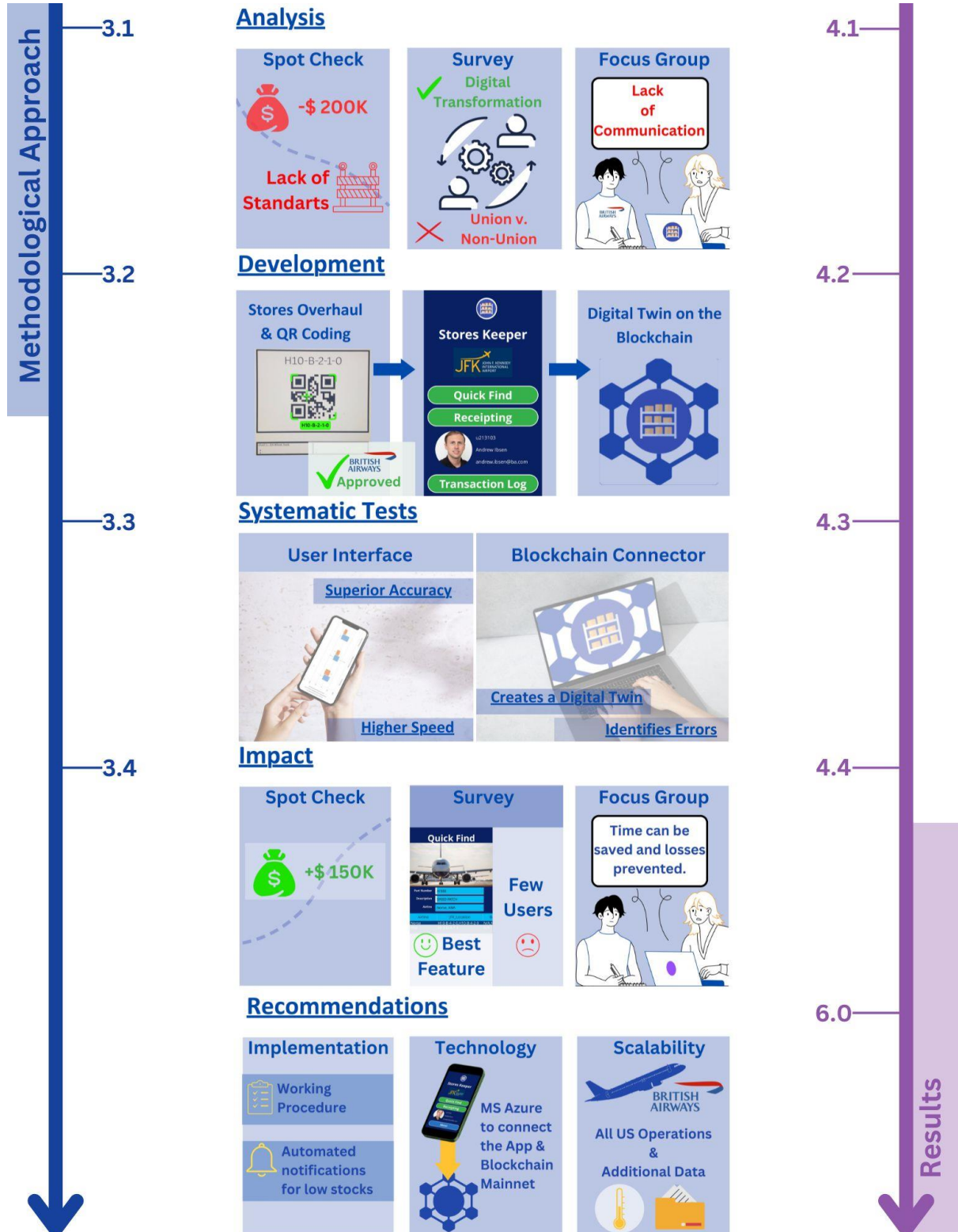
Overall, the project has successfully implemented and tested a tokenization of parts for inventory management in line maintenance. It has done so by harnessing publicly available tools, exploiting complex technology with little programming and software development expertise, at minute cost and in compliance with cyber-security standards. Thus, the much-discussed trend of blockchain for inventory management in aviation was put to test. Its value was proven, provided it is built upon real-life, bottom-up meticulous organization of stores and their information systems, tailored to the operation’s capabilities and personnel.

6 Conclusion

6.1 Achieving the study's objectives

The results outlined offer a compelling narrative on the transformative potential of blockchain technology in aviation maintenance. They provide a roadmap for future enhancements and broader implications for the aviation industry. All 4 main objectives were met as listed and illustrated below, in Figure 24 .

1. Main strengths and limitations of spares management at BA's JFK station at the onset of the project included:
 - a. A spot-check detected \$200K in misplaced parts, while a systematic analysis of the system found a lack of standards in inventory-keeping.
 - b. A survey of users uncovered a rift between unionized and non-unionized employees as to best practices in inventory-keeping, but an overall willingness of all for digital transformation.
 - c. A focus group and semi-structured interviews with key stakeholders brought out a lack of effective communication regarding stores.
2. Building of a user-friendly mobile application to tokenize spares comprised:
 - a. A hierarchical location system with QR codes was implemented in the physical stores.
 - b. A QR-scanning MS Power inventory management mobile application, called Stores Keeper App, was built.
 - c. A decentralized application that interacts with a blockchain, tokenizing spares, was developed. Together with the mobile application, it constitutes the Stores Keeper Solution.
3. Test the viability of tokenizing spare parts in BA's hangars included:
 - a. An experiment with the mobile application found it to improve accuracy and time taken to complete transactions in and out of stores.
 - b. A set of simulations with the decentralized application found it to be success in preventing errors in stock keeping and in keeping an immutable digital twin of stores record.
4. Qualify and quantify the impact of the proposed solution to tokenize spare parts.
 - a. A new spot-check found 50K in misplaced parts, or a 150K cost-saving relative to the spot-check at the onset of the project (1.a).
 - b. A survey of users rated the application positively overall, despite few new users having been acquired.
 - c. A focus group and targeted interviews highlighted the potential time and cost saved thanks to the solution.



6.2 Additional take-aways

In the dynamic environment of line maintenance, having centralized access to part availability and the capabilities, in both hardware and software, to transact components quickly and efficiently encourages the user to not cut corners – the path of least resistance. Here, implementing complex technology such as a distributed ledger proves to be effective only when stores are re-organized from the bottom-up. Furthermore, the Stores Keeper creation process highlights the importance of tailor-made solutions, based on in-depth analysis of the systems and information flows in place. In addition, involving future users from the project's onset proved to be extremely valuable for insight, feedback and, ultimately, adoption of the tool created. Since efficient technology is available in low-code and AI assisted platforms such as MS Power Apps, effort can move away from programming challenges, instead going further into refining understanding of needs, translating into better design. Lastly, even efficient, and well thought-out solutions do not exist in a void, and must be integrated with larger, possibly outdated, systems. Thus, it is of the essence to also work closely with IT departments, to create customized and secure integration solutions as legacy systems are gradually upgraded.

Next steps to further enhance the Stores Keeper solution include deploying the smart contract on a "mainnet" (\$70.00) and running blockchain transactions on it (projected \$10.00/year) as well as integrating the mobile and decentralized applications creating using a custom connector on MS Azure (estimated at \$ 100.00/year). It also includes integrating the system to SAP records using GUI Scripting once the SAP system is updated, in April 2024. Furthermore, scaling to all stations in the US is envisioned and supported by BA. With this, it would be conceivable to move away from external blockchain and implement a proprietary blockchain, distributed across BA stations. Another potential add-on currently being considered is using computer vision tools to digitize part's documentation when transacting them into the stores. This would prevent storage of parts without proper document checks (see diagram in 4.1.1) and allow for more comprehensive audits to be conducted. In the same spirit, a temperature monitoring system is being installed in JFK stores, using a miniature Raspberry Pi computer (\$ 70), with temperature records stored in a blockchain. This allows for the retrieval of continuous and tamper-proof graphs of the store's conditions for audits and, most importantly, to trigger immediate email alerts to personnel in charge when regulations are breached. Successful implementation in the scope of this project has paved the way to building custom application, bringing BA stores to the forefront of technology and, most importantly, making much-needed strides in efficiency and safety.

6.3 Avenues for research

Thanks to a system such as the Stores Keeper, data is systematically collected and made tamper-proof. This data can now be analyzed to increase efficiency of inventory management, procurement, and overall strategy. It has been shown across industries that inefficient inventory management can lead to financial loss and delays, with the main risks to the companies being financial, i.e. loss of capital, operational, i.e. poor forecasting and

planning, and strategic, i.e. inappropriate decision-making on pricing and diversification (Disk, 2023). For airline's operational cost, stores that are not optimized lead to reactive rather than proactive maintenance strategies. This approach tends to be more costly due to the need for urgent repairs and procurement of items at short notice. Inefficient inventory management can strain relationships with suppliers and disrupt the supply chain, leading to challenges in obtaining parts in the future. It is thus of the essence not only to correctly monitor levels in stores but also to predict future needs before conflicts arise. A growing research topic in this domain is thus predictive forecasting (Daily and Peterson, 2017; Gilges, 2023), powered by machine-learning technology, neural models and, more recently, large language models (LLMs), which are by definition predictive models for sequential data. The promise of LLMs is further enhanced by the very recent development of long-term memory mechanisms such as MemoryBank, that would allow them to further tailor responses to users (Zhong et al., 2023). Beyond theoretical research, such models must now be applied and tested in the field.

In addition, the data collection could be expanded to include personnel time on aircraft, personnel present at the station across teams, use of tools and vehicles, flight turnaround and other crucial variables, which can all be thought of as, essentially, transactions. Such large data sets could then be explored with predictive modeling tools. The company Moonware, for instance, inspired by Uber, explores vehicle tracking and ground traffic control to streamline turnaround performance. It has received \$ 2.5 million in pre-seed financing (Alamalhodaie, 2023). Preventing inefficiencies in labor and resource allocations is the overarching theme. In addition to their potential benefit for the airline's balance sheet, such methods would enhance safety. In fact, models could be geared towards detecting the provenance of undocumented parts, the cause for repeated technical errors and other pitfalls, allowing for prevention to be implemented. Furthermore, digitizing documentation upon receipt and systematically recording counterfeits would allow for AI models to be trained in recognizing such counterfeits and, ultimately, emitting warnings in real-time. Technology is available to accurately collect and process large data sets continuously, building and training machine learning models in-house for many of airline's everyday engineering operations and challenges.

When it comes to strategic planning, if costs of maintenance are poorly accounted for, because the turnover of parts is not routinely monitored, decisions of fleet updates and routes are ill-informed. Strategic risks are also relevant in complex maintenance operations, such as in airlines selling engineering as a service as a secondary business such as at BA's JFK operation. This fairly common practice allows carriers to have access to engineering outside of their main hubs with fewer overhead. However, it requires that the provider and the customer airline work together in keeping optimal stores outside of their territory, although paradoxically under the remote national regulations of the foreign customer airline. In such settings, the decision of taking on new customer airlines by the service provider needs to be carefully weighed against personnel, equipment, and processes available, and priced in accordance. Overall, poor part control is linked to significant financial, operational, and

strategic mistakes that impact a carrier's balance sheet and survival in a competitive market. Having access to continuous quality data through a solution like Stores Keeper allows the creation of models and implementation of simulations that can effectively guide strategic solutions. As for inventory keeping, accurate and large data collection is the essential first step when building and training machine learning models.

Overall, this project and its future developments showcase that powerful in-house technological solutions can be created organically, by personnel on the ground with only average digital literacy. Having such bottom-up digital creation leads to continuous data collection that is well integrated with operations, meaningful and secure, as it is both encrypted and made tamper-proof by the blockchain. This is all the most important considering that predictive AI tools, the natural next step to guide airlines' management, strategy, and safety procedures, will only ever be as good as the data they are trained and tested on. Taken together, these elements bode a strategic question for airlines, one that warrants future investigations. That is, should airlines operate with a technology-forward mindset, giving training, tools, and technical support to employees at all levels to independently automate day-to-day operations? A structured system, allowing for pilot-testing of applications thus created and cyber security risk assessments, coupled with rewards for employees taking initiative could be imagined. Such a logic could safely launch an airline to the forefront of the technological transition we are living in all the while solving long-standing vulnerabilities. Furthermore, in the engineering department, recruiting geared towards individuals willing to engage with digital tools will build a workforce that is better suited to learn and care for ever-more digitally intensive aircraft. A bottom-up data-driven approach can result in a gain in efficiency and reliability, ultimately moving the whole industry towards higher safety standards.

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